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Generative AI has the potential to transform productivity and reduce inequality, but only if adopted broadly. In this paper, we synthesize evidence from 76 sources from over 100 countries and analyze web traffic data to show that gender gaps in generative AI use are nearly universal and persistent. Focusing on the 318,924 respondents from sources that report usage rates for both men and women, we estimate an AI adoption rate of 47.8% for men versus 39.3% for women, a relative gap of 22%. While this relative gap has shrunk over time, it has stabilized at roughly 16% since early 2025. These patterns hold in global web traffic data on the ten most visited AI tools. This data also reveals that women generally spend less time using AI and that the gaps are largest for frontier tools. Our findings are consistent with a view that as AI diffuses, the gap shrinks as women are exposed to and gain familiarity with AI, but that institutional, organizational, and social frictions will likely lead to persistent gaps in who uses AI.

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# 1 Introduction

Generative AI is expected to have profound economic and social impacts. Recent studies demonstrate that AI can already impact the skills, knowledge, and productivity of professionals across various domains, including college-educated workers, customer support agents, job seekers, students, and entrepreneurs (Brynjolfsson, Li, and Raymond, 2025; Dell'Acqua et al., 2023; Noy and Zhang, 2023; Kim, Kim, and Koning, 2026). Moreover, because these tools are generally both widely accessible and easy to use, they have the potential to help billions of people from historically underserved groups across the world (Björkegren, 2023; Otis et al., 2023). This is especially true for women, who still face widespread institutional, career, and cultural barriers that prevent them from gaining the skills and knowledge needed to succeed at work (England, Levine, and Mishel, 2020a; Goldin, 2021). If AI lowers the barriers to gaining the skills and knowledge required to be productive at work, then these tools might especially benefit women.

Yet the history of technology adoption suggests women may use AI less than men. Because of gendered occupational segregation, men and women often work in different firms, jobs, and occupations, thus exposing women to fewer opportunities to encounter certain technologies at work. This is reinforced by gendered networks, limiting how AI tools and know-how diffuse to women (DiMaggio and Garip, 2012). Even when women have access, stereotypes and status beliefs may hold them back: using tools like AI could carry a greater risk of being penalized or seen as "cheating" for women than for men (Correll and Ridgeway, 2003; Ridgeway, 2011; Carvajal, Franco, and Isaksson, 2024). Recent evidence has also found that women face higher perceived and actual exposure to AI-driven job displacement, which dampens enthusiasm for adopting tools with the potential to reshape their work (Borwein et al., 2026; Lean In, 2026). Compounding all of this, the systems themselves are trained on biased data (Guilbeault et al., 2024), including data from their predominantly male users (Cao, Koning, and Nanda, 2024), so the tools women do reach may well work less well for them.

In this paper, we show that there is a large, persistent, and nearly universal gender gap in both the adoption and the intensity of generative AI use. We do so by first assembling evidence from 76 academic, industry, and government sources on gender and generative-AI use, of which 58 directly report or allow us to estimate adoption rates for men and women. Our results reveal a remarkably consistent gap: across all sources that report adoption rates and sample sizes split by gender, men's adoption is roughly 22% higher than women's. We investigate how the gender gap in AI usage has changed over time, from just after the launch of ChatGPT in 2023 through early 2026. We show that while the gap has narrowed over time, it remains significant, and appears to have stabilized, with men approximately 16% more likely to use AI compared to women since 2025.[1] We show that

[1] This estimate uses the 2025–2026 systematic review sources that have comparable adoption rates and available sample

these gaps cannot be fully explained by occupational segregation, with many sources showing that even in the same occupation or firm, women use AI less than men. We also show that these gender gaps in AI usage exist both inside and outside of work settings.

Second, we analyze web analytics data from SimilarWeb on the gender composition of traffic to the ten most visited AI websites. Consistent with the findings from our systematic review, we show that while the share of women using AI tools has increased, women still make up the minority of traffic to these tools, with especially large gaps in India, Pakistan, Iraq, and South Korea. We further document that these usage gaps are smallest for companion-oriented chatbots and largest for newer and more technical tools, such as vibe coding app builders. Even among women who use this technology, we also find women engage less frequently and for shorter durations than men.

Third and finally, we organize the underlying mechanisms explored in the 76papers included in our systematic review. We find that these mechanisms can be roughly grouped into two sets. First, there is evidence that there are simple differences rooted in knowledge and familiarity with AI tools, differences that the market is likely to close as these tools diffuse. The second set of mechanisms focus on institutional, social, and behavioral differences that might drive more persistent differences in usage. Our analyses show that the gap is shrinking, but persistent, suggesting that both sets of mechanisms are at work.

Our paper makes two broad contributions. First, we provide much needed systematic evidence on gender and generative AI use across the globe. We provide a map of which populations and tools have been studied and how gender gaps have been changing over time. This map helps rationalize recent estimates showing the gap may have closed for some types of users and tools (Chatterji et al. (2025)), but not others. Taken together, our findings show that even in 2026, there are meaningful gender gaps in AI usage and that researchers should continue to explore the drivers of these gaps and how to close them.

Turning to our second contribution, our paper helps organize the potential underlying mechanisms behind these disparities that future researchers should investigate. While there are clearly still knowledge and familiarity gaps—especially for frontier models and tools—our analyses point to the need to understand the institutional, organizational, and behavioral mechanisms that lead to persistent differences in usage. Whether generative AI ends up narrowing or widening existing gender inequalities will depend on whether these frictions are addressed directly.

sizes. We compute sample-size-weighted adoption rates separately for men and women as $\bar{m}_w = \sum_i N_i m_i / \sum_i N_i$ and $\bar{f}_w = \sum_i N_i f_i / \sum_i N_i$, where $m_i$ and $f_i$ are the male and female adoption rates in source $i$ and $N_i$ is the source's analytic sample size. Across the 16 qualifying source-outcome entries ($\sum_i N_i = 113{,}981$), $\bar{m}_w = 39.8\%$ and $\bar{f}_w = 34.4\%$, so the absolute gap is 5.4 percentage points and the relative gap is $(\bar{m}_w - \bar{f}_w)/\bar{f}_w \approx 15.7\%$.

## 2 How Gender Relates to the Adoption of Generative AI

Generative AI has generated extensive excitement for its potential to increase productivity. Recent evidence has shown that it can help with writing (Noy and Zhang, 2023), raise productivity in customer-support work (Brynjolfsson, Li, and Raymond, 2025), and can help people learn new skills, such as helping consultants engage in data science tasks (Wiles et al., 2026). In line with this excitement, Autor (2024) argues that AI could reduce inequality by extending expert decision-making to a broader population of workers. However, the effects of generative AI may be uneven and amplify existing (dis)advantages (Otis et al., 2023; Dell'Acqua et al., 2023). Understanding who benefits from generative AI is crucial to understanding its productivity gains at scale. In this paper, we focus on how gender relates to the adoption of generative AI.

Why might men and women adopt and use generative AI differently? Given the scale and persistence of gender differences in the workplace and society, as well as historical technology adoption patterns, gender represents a salient lens for understanding AI diffusion (England, 2010; England, Levine, and Mishel, 2020b). A number of potential mechanisms could explain why women and men may adopt and use generative AI differently. We focus on two main classes of mechanisms that have different implications for the potential persistence of this gap over time. The first class of mechanisms relates to information lags, such as knowledge about and familiarity with particular tools, that may fade over time. The second class of mechanisms are related to persistent behavioral and organizational factors that are unlikely to change rapidly. In this section, we focus on both classes of mechanisms and their implications for whether gender gaps would improve or self-correct over time.

The first broad class of mechanisms relates to knowledge about, familiarity with, and perceived usefulness of AI. Women may have less knowledge about or familiar with generative AI, which could be due to occupational sorting (Reskin, 1993; England, 2010; Goldin, 2014). Men disproportionately hold technical and managerial roles where generative AI is most likely to be deployed (Baron and Pfeffer, 1994; Petersen and Morgan, 1995; Tomaskovic-Devey and Skaggs, 2002). Such gender differences in occupations are due to a wide variety of norms and factors, for example, women are more likely than men to leave engineering majors (Cech et al., 2011). In this context of gendered occupational segregation, men may be more likely to adopt generative AI early if they benefit from awareness and observability. This may be especially likely given that workers often learn about tools and how to optimally use them from their peers and networks (Rogers, Singhal, and Quinlan, 2019; DiMaggio and Garip, 2012). Historically, this pattern has occurred for various technologies, including the internet (Bimber, 2000). In line with gender occupational segregation, generative AI likely may not feel as useful to the kind of work that women do, or women may not be as immediately concerned with or exposed to the technology. Alongside these explanations, we may expect to see differential adoption across countries and locations, based on differences in gender norms. If this set of explanations is what is driving the gap, then with time, the technology will eventually diffuse, and the market can "solve" this gap. Thus, we should expect the gender gap to close over time. Moreover, even if most of the gap is driven by differential diffusion, generative AI is a setting of incredible growth and rapid innovation, with new technologies and frontier tools emerging continually (Maslej et al., 2024). Therefore, even if tools eventually end up diffusing broadly, the pattern of differential diffusion may repeat itself for every new wave of AI technology, which could lead to persistent gender gaps. Indeed, a recent large-scale survey has shown that male social scientists are much more likely than women to adopt frontier agentic coding tools such as Claude Code and Codex (Lyttelton, Massenkoff, and Wilmers, 2026).

However, a second class of mechanisms may be at play that would have different policy and managerial implications. This set of mechanisms includes gender differences in confidence, competence penalties, as well as trust, risk, and norms. If these mechanisms dominate, without changing tools or organizations, the gender gaps in AI usage would likely persist over time.

One potential mechanism is reduced confidence in working with AI. Women consistently report lower self-assessments of technical competence, even when objective performance is equivalent (Correll, 2001, 2004). Women also tend to engage in self-stereotyping and are less willing to contribute ideas in counter-stereotypical domains, even holding ability constant (Coffman, 2014). Women also tend to engage in less self-promotion than men (Exley and Kessler, 2022). Because the effective use of AI requires iterative and exploratory engagement, like prompting, evaluating, or re-prompting, the process of using AI rewards the kind of self-confident experimentation that gender norms have historically suppressed in women.

Women may also use AI less because they receive less credit for their work. In professional collaborative settings, women's contributions are systematically undervalued and less often attributed to them than men's equivalent contributions (Sarsons, 2017; Castilla, 2008). This attribution asymmetry could weaken the career incentive to master a tool whose payoff depends on visible recognition. At the same time, if AI assistance is interpreted as a signal of lower ability, women may also be more apprehensive about using AI because they fear appearing incompetent for doing so. If women are held to a higher standard to prove their competence, the same AI-assisted output is more likely to be read as a sign of low ability for a woman than for a man, raising its reputational cost (Ridgeway and Correll, 2004; Biernat and Manis, 1994).

Another reason why women may persistently lag men in AI adoption is the lack of training and institutional support. Using AI effectively often requires substantial upfront investment, including learning effective prompting strategies, building prompt libraries, and integrating tools into workflows. Women's disproportionate responsibilities for unpaid domestic and caregiving labor may leave less discretionary time available for this investment (Bianchi et al., 2012; Delecourt and Fitzpatrick, 2021; Goldin, 2014), especially when payoffs are uncertain. Without dedicated training and institutional support to offset this cost, women may be more likely to under-invest in AI relative to men.

Finally, women may have different concerns about AI than men, especially concerns related to trust, privacy, and risk. Women tend to be more risk-averse than men (Croson and Gneezy, 2009). They may also be more sensitive to the gender biases online that become embedded in generative AI tools (Guilbeault et al., 2024; Guilbeault, Delecourt, and Desikan, 2025; Kirk et al., 2021). Together, this class of mechanisms suggests that gender gaps in generative AI adoption may not simply reflect information lag, but rather behavioral and organizational dynamics that are deeply entrenched in society.

Overall, these two sets of mechanisms have different implications regarding whether gender gaps would persist over time. Yet, even if the gap ends up closing, any gap in usage may have lasting implications. Cao, Koning, and Nanda (2024) has shown that early users shape tools, which could make these tools more appropriate, useful, or tailored to men's usage. In addition, early users may get premiums from using a new technology (Barley, 1986; Orlikowski, 1992). First movers might be able to capture resource and reputational advantages that are difficult for later entrants to overcome (Lieberman and Montgomery, 1988). Finally, consistent with the Matthew effect, early differences in familiarity, confidence, or institutional endorsement may compound over time as early users accumulate experience, productivity gains, and social legitimacy (Merton, 1968). In other words, the timing and sequence of adoption may still matter even if the gender gap ends up eventually closing.

# 3 Data and Methods

We combine two primary data sources to characterize the AI gender gap. First, we compile a systematic review of over 70 studies that measure AI use by men and women. Second, we build a monthly panel that tracks visits by men and women to the most popular consumer facing AI tools, allowing us to measure usage over time, across countries, and between products.

## 3.1 Systematic Review

To understand the AI gender gap, we conducted a systematic review of the empirical literature by identifying any paper that reported AI usage rates split out by men and women. As of May 28th, 2026, we have identified 76 sources. Of these, 18 are discussed in the following narrative review but are not numerically summarized because they do not report directly comparable male and female adoption rates for the same AI-use outcome. Turning to the 58 studies that report the share of women and of men who each are using AI, 54 of which also report the exact count of men and women in the data (i.e., the denominator). The result of our systematic review is a sample of 318,924 people for which we can know their gender and AI use. All our sources are summarized in Table A3 and Appendix A describes our search strategy, eligibility criteria, source list, and coding decisions in further detail.

Our sources differ in sampling frame (including nationally representative adult panels, workforce panels, higher-education samples, and single-firm traces), definition of AI use (including ever use, weekly use, workplace vs personal use, and tool-specific use), and geographic coverage, spanning more than one hundred countries. Our review includes groups as varied as Danish knowledge workers (Humlum and Vestergaard, 2025), medical clinicians in Switzerland (Egli et al., 2025), journalists in the United Kingdom (Thurman, Thäsler-Kordonouri, and Fletcher, 2025), medical students in Sudan (Ahmed et al., 2025), full-time software engineers at a global technology company (Gai, Hou, and Tu, 2025), natural- and applied-science researchers in Germany (Chugunova et al., 2026), higher-education students in Ghana (Ofosu-Ampong, 2023), a representative sample of Korean workers (Suh and Oh, 2026), and Norwegian business students (Carvajal, Franco, and Isaksson, 2024).

### 3.2 Web Analytics Data on AI Tool Use

Survey adoption rates are informative but, like all measures, have limitations. It's possible that respondents misreport and samples skew toward easy-to-reach populations. Few studies track usage over time with consistent samples or across more than a handful of countries. To complement our systematic review, we therefore turn to web analytics data that tracks the visitors to the web versions of the most popular AI tools overtime. Specifically, we use data from SimilarWeb, a digital-intelligence vendor that estimates website-level audience composition by combining a global panel of users from anonymized clickstream traces, ISP-level data, and public web signals. For each domain-month, SimilarWeb reports total visits and breaks the audience down by gender and age band. Our approach builds on prior research using Similiarweb to measure startup performance and demographic gaps (Koning, Hasan, and Chatterji, 2022; Cao, Koning, and Nanda, 2024).

Using this data we build a panel that tracks the gender composition of the ten highest-traffic consumer facing AI websites worldwide, monthly, from January 2023 through January 2026. We rank domains in SimilarWeb's AI Chatbots and Tools category by average monthly visits over the most recent six-month window (August 2025 through January 2026) and keep the top ten:

*chatgpt.com*, *gemini.google.com*, *chat.deepseek.com*, *grok.com*, *claude.ai*, *character.ai*, *perplexity.ai*, *copilot.microsoft.com*, *polybuzz.ai*, and *chat.qwen.ai*. Together these tools account for roughly 98% of all visits in the AI category over that window.[2] Where a single product spans multiple domains, we pool them using traffic-weighted demographic shares.[3] The result is a panel that captures the bulk of consumer-facing AI traffic and lets us track gender composition overtime. Our primary outcome is the percent of total traffic share that can be attributed to women. We use this worldwide top-ten panel to examine the average female share of AI-tool website traffic over time and to compute country-level averages for 59 countries from August 2025 to January 2026.

We also build a specific panel specific to the United States that tracks the gender composition of the ten highest-traffic consumer facing AI websites in the US, again from January 2023 through January 2026. As before, we rank domains by average monthly U.S. visits from August 2025 through January 2026 and keep the top ten: *chatgpt.com*, *gemini.google.com*, *grok.com*, *claude.ai*, *character.ai*, *perplexity.ai*, *copilot.microsoft.com*, *chat.deepseek.com*, *polybuzz.ai*, and *base44.com*. Together they account for roughly 98.2% of all U.S. visits in the panel over that window.[4]

# 4 Results

## 4.1 Systematic Review: Gaps and Trends

In Figure 1, we show adoption rates across the 58 sources in our systematic review that report sample sizes and usage rates split by gender. Specifically, we plot the percentage of men compared to the percentage of women who use AI in each source. Across the 58 plotted sources, the raw gender gap is positive in 56 sources and negative in 2. Pooling across the 318,924 respondents or observed users in the 54 figure sources, sample-size-weighted men's adoption is 47.8% and women's is 39.3%, a gap of 8.5 percentage points. In relative terms, men are about 22% more likely than women to report using AI.[5] The spread across sources is wide, ranging from a 23.6-percentage-point

[2]Share computed from SimilarWeb's category-level extract for March 2026, which lists all 70 indexed domains with their worldwide monthly visits. The next two domains, *kimi.com* and *lovable.dev*, each account for less than half a percent of category traffic.

[3]Consolidated groupings: *chatgpt.com* absorbs *chat.openai.com* and *openai.com*; *grok.com* absorbs *x.ai*; *claude.ai* absorbs *anthropic.com*. Weights are monthly all-traffic visits from SimilarWeb.

[4]Share computed from the current 53-domain SimilarWeb panel, using average monthly U.S. visits from August 2025 through January 2026. The next two domains, *lovable.dev* and *meta.ai*, each account for less than 0.3% of U.S. panel traffic.

[5]We compute sample-size-weighted adoption rates as weighted averages of the source-level male and female adoption rates, using each source's total sample size as the weight. Public First UK Workers and Eurostat are plotted but excluded from the weighted calculations because they do not report numerical sample sizes for the plotted comparison. Salesforce/YouGov and ZipRecruiter are also excluded because they report approximate, rather than exact, sample sizes. The weighted sample includes 54 source-outcome entries with exact numeric sample sizes, covering 318,924 respondents or observed users. This yields adoption rates of 47.8% for men and 39.3% for women, an 8.5-percentage-point absolute gap and a 21.7% relative gap. Unweighted averages across the same entries are 54.2% for men and 44.4% for women, implying a 8.5-point absolute gap and a 21.7% relative gap.

gap favoring men to a 3.0-percentage-point gap favoring women. Table A1 lists each source with its sample description, source type, and data-collection period. Table A2 reports the exact survey-question wording (where available) for each estimate, and otherwise gives the closest published description of the plotted measure.

Three sources run counter to the main pattern we find of women using AI less than men. A survey conducted by Boston Consulting Group (BCG) in January 2024 found that female tech workers were 3% more likely to use generative AI than men in a convenience sample of 6,558 employees (Barisano et al., 2024). This positive gap is driven primarily by women working in senior technical roles, whereas junior women are significantly less likely to use generative AI than junior men (Barisano et al., 2024). The second finding appears in Alonso-Muñoz and Casero-Ripollés (2026), whose online survey in October 2025 of 281 journalism students at a Spanish university reports that 94.48% of female students and 91.53% of male students used generative AI. Finally, Shishakly (2025) surveys 506 higher-education students in November 2024 in the United Arab Emirates and finds that female students used ChatGPT more frequently than male students.[6] However, across the rest of the sources in our systematic review, women are much less likely to use generative AI than men.

One concern about these results is that these gender gaps are merely due to occupational sorting, e.g., men may be more likely to work in technical or analytical occupations where generative AI is especially useful and encouraged. However, many sources in our systematic review show that gender gaps in AI usage hold within the same occupation, industry, and organization. For example, Gai, Hou, and Tu (2025) track full-time software engineers at a single global technology company following the launch of an internal AI coding tool in early 2024 and find that 43% of male engineers but only 31% of female engineers used the tool at least once, despite identical job roles and identical access. We also see this pattern across other professional populations: among Swiss clinicians, Egli et al. (2025) find that 37.4% of men but only 26.5% of women are frequent users of large language models in late 2024; among a representative sample of UK journalists, Thurman, Thäsler-Kordonouri, and Fletcher (2025) report that 36% of men versus 30% of women use AI professionally at least once a week in late 2024; and among publishing academics surveyed across twenty countries, Mohammadi et al. (2026) find that female researchers are roughly ten percentage points less likely than their male counterparts to use AI frequently for research in early 2024. Similar within-institution gaps appear in education settings: at the Norwegian School of Economics, Carvajal, Franco, and Isaksson (2024) document that 75.7% of male students versus 60.7% of female students report regular generative AI use in late 2024, while at a US liberal arts college, Contractor and Reyes (2025) find that 88.7% of male undergraduates versus 78.4% of female

[6] Because the paper reports only the overall mean AI use on a 5-point Likert scale and a qualitative gender comparison, this source could not be pooled with the binary adoption rates used in our figures.

undergraduates have ever used generative AI for academic purposes in early 2025. Most comprehensively, Humlum and Vestergaard (2025) study a representative sample of 18,000 Danish workers in early 2024 drawn from eleven specific AI-exposed occupations (including accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketers, office clerks, software developers, and teachers) and show that women are 16 percentage points less likely than men in the same occupation to have used ChatGPT for work. Most recently, in a February-March 2026 study of quantitative social scientists (including economists, psychologists, sociologists, and political scientists), researchers at Anthropic found that 81% of respondents with typically male names and 78% of respondents with typically female names had previously used generative AI models to aid their research process (Lyttelton, Massenkoff, and Wilmers, 2026). Taken together, these within-occupation and within-institution comparisons indicate that the gap cannot be fully explained by sectoral segregation of men and women across the labor market.

The gap also extends beyond the workplace. If it were driven entirely by occupational tasks, employer policies, or exposure through jobs, we would expect personal-use gaps to be small or non-existent. Yet, we see gender gaps in both professional and personal AI use. For AI use at work, using nationally representative U.S. data in late 2024, Bick, Blandin, and Deming (2024) find that 29.1% of employed men versus 23.5% of employed women used generative AI for their job, and Lean In (2026) reports similar workplace gaps. In a February 2026 survey of US job seekers, Bachaud (2026) find that 65.6% of men versus 58.5% of women have used AI in their job search, including for interview preparation, drafting resumes and cover letters, searching for jobs, improving LinkedIn profiles, and analyzing job descriptions. Outside of work, English et al. (2026) document that men are more likely than women to use AI for leisure and personal activities in France, Spain, Poland, the United States, Canada, and Australia in late 2025. Evidence that compares gender gaps in work and personal use directly is mixed: Howington (2024) finds a larger work-related gap among U.S. working professionals compared to personal use in early 2023, while Stephany and Duszynski (2026) finds a larger personal-use gap using nationally representative data from the UK in mid 2024. Either way, gaps appear even outside workplace settings, where employer policies, occupational tasks, and formal access should matter less.

Beyond shedding light on how the gap varies across occupations, industries, and organizations, our systematic review also allows us to explore how the gap has evolved over time. Figure 2 plots the gender gap in each figure source against month the data was collected, specifically the difference between the share of men that report using AI minus the share of women that report using AI.[7] In 2023, the first year after the launch of ChatGPT, the gender gap in AI usage ranges from 5 percentage points to nearly 25. Consistent with the idea that women are slower to adopt new

[7] When data collection spanned multiple months we use the last month that was collected.

technologies ((Bimber, 2000)), we see a marked decrease in the number of studies that report gaps over 10 percentage points in more recent months. However, there does appear to be a persistent difference in usage, with the gaps in March 2026 spanning from around 3 to 13 percentage points, with the 13 percentage point gap coming from differences in frontier agentic coding tools (Lyttelton, Massenkoff, and Wilmers, 2026). This suggests that even as overall usage gaps shrink, as new more powerful AI tools come to market, women are likely to remain behind the frontier.

### 4.2 Web Analytics Results

Our analysis above shows that women self-report using AI at lower rates than men, and that this pattern holds across occupations and time. To further unpack this pattern, we turn to web analytics data that allow us to track trends by month, country, and tool, providing finer grained analyses and allowing us to address potential concerns around publication bias. Published studies may disproportionately study early adoption, the U.S. or high-income western contexts where norms about and access to AI may differ from the rest of the world. Using monthly web traffic data allow us to see whether the gender gap persists beyond the earliest stages of adoption, and country-level data allow us to examine whether similar gaps appear in places that are currently underrepresented in the systematic review. Finally, while surveys tend to focus on whether someone uses AI at all, our web analytics data lets us observe which tools men and women use and how intensively they use them.

In Figure 3, we plot the average share of worldwide AI-tool website traffic coming from women among the 10 most popular AI websites from January 2023 to January 2026 across the globe. The red line shows the cross-tool average in each month, and the shaded band shows the cross-tool 95% confidence interval. Across the ten highest-traffic AI tools, which account for nearly all traffic in SimilarWeb's AI Chatbots and Tools panel during the last year, we find that women are almost always in the minority of visitors to these websites. Consistent with Figure 2, we see the gap narrows over time. In Jan 2023, just under 35% of visitors to these AI websites were women. By January 2026, about 40% were women. Looking at the figure, while the gap has clearly shrunk, it appears to have been stable at 40% since October 2024, suggesting that it is unlikely to close in the near future.

We next turn to geographic variation in gender gaps. In Figure 4 we plot the female share by country using visit data from August 2025 through January 2026. In the vast majority of countries, women account for less than half of observed traffic, although the size of the gap varies substantially across locations. The largest gaps appear in Pakistan, India, Iraq, and South Korea, where women's share of observed AI-tool visits remain below 35%. These patterns are generally consistent with differences in gender norms, labor market participation, and access to digital technologies across countries. On the other hand, the countries that appear closest to gender parity include countries like Hungary, Japan, Ukraine, Hong Kong, and New Zealand. However, this data should be interpreted with some caution. Many of these estimates from smaller countries have smaller traffic volume and therefore have noisier gender estimates. That said, the country-level gaps remain quite large, in many cases remaining larger than the early gaps seen in the US.

Which types of AI tools are men and women most likely to use? In contrast to survey data which tends to focus on whether respondents have ever used Generative AI tools, our analytics data allow us to measure the share of traffic data to a specific tool coming from women vs men. Focusing on the United States, in Figure 5, we show how the female share of U.S. traffic varies across tools over time.[8] Table 1 summarizes the same U.S. top-ten tool panel over August 2025–January 2026, ranking tools by average monthly U.S. visits and reporting gender composition and visit duration. Focusing first on traffic shares, all ten tools are male-skewed, with female shares ranging from 26.7% on `grok.com` to 44.1% on `chatgpt.com`. These gender gaps are statistically significant for every tool. The largest tools are also male-skewed: among `chatgpt.com`, `gemini.google.com`, `grok.com`, `claude.ai`, and `character.ai`, which together account for 92.4% of average monthly U.S. panel traffic, none reach gender parity. This tool-level pattern appears in both mainstream general-purpose tools and more specialized or newer AI interfaces, such as `base44.com`, a vibe coding app building platform. We also see this pattern of especially large gender gaps on newer tools in our systematic review. In a recent report by Anthropic, Lyttelton, Massenkoff, and Wilmers (2026) found that quantitative social scientists with more typically male names are 144% more likely to use coding agents than those with more typically female names more than once a week. [9] Appendix B shows that this product-level heterogeneity extends to AI tools outside of the 10 most popular on SimilarWeb. Using data from *There's An AI For That*, an online directory that catalogs specialized AI tools by use case and links to each tool's website, we find that women are relatively more represented on tools for education, language learning, and personal-life assistance, while men are more represented on tools for coding, website building, finance, analytics, and other technical or productivity-oriented tasks.

Our tool-level analysis also motivates a closer look at `chatgpt.com`. ChatGPT is the dominant platform in the U.S. panel, accounting for over 70% of average monthly U.S. AI tool traffic. It is also the only platform in our panel for which the provider has released any internal data on demographic user composition. Chatterji et al. (2025), using a uniform sample of 1.1 million consumer ChatGPT accounts from across the globe, show that the platform's initial male skew has narrowed

[8] We focus on the United States as traffic volumes are large enough to compare products and usage intensity within one context. Especially given that individual tools are not necessarily available across all countries in our sample, focusing on the US ensures any differences are from who uses the tool in a given country rather than differences in which countries can use each tool.

[9] 22% of respondents with typically male names reported using these tools more than once a week, while only 9% of respondents with typically female names reported using these tools more than once a week. Thus, $\frac{22-9}{9} \times 100 = 144.4\%$. Coding agents include Claude Code, Cursor, Codex, and Google Antigravity.

over time. Among users who are active during a week and whose first names can be classified as typically masculine or feminine, they show that the share with typically feminine names reaches near parity, and even slightly exceeds parity, by mid-2025.

At first glance, this result may seem to conflict with both the survey evidence above and our SimilarWeb estimates of the gender gap in AI usage. In our SimilarWeb data, women account for only 44.1% of US `chatgpt.com` website visits over August 2025–January 2026 (Table 1). The U.S. time series in Figure 5 shows `chatgpt.com` remaining below parity throughout the period, hovering around 44% female in recent months. Looking at our global data, we similarly see that while `chatgpt.com` is closer to gender parity than most AI tools, women still make up approximately 45% of traffic visitors. However, these estimates reflect different populations. Chatterji et al. (2025) sample includes only users on consumer accounts who are active in a given week. It does not include users who are using ChatGPT without an account or who are using ChatGPT through a business, enterprise, or education plan. By contrast, our SimilarWeb estimates measure total website traffic from all users, no matter their account status, which provides a potential explanation for the differing estimates.

Why might restricting the sample to users with consumer accounts lead to different estimates? Suppose men are more likely than women to have business or enterprise accounts–potentially reflecting higher workforce participation and sorting into AI-intensive occupations. They would be missing from Chatterji et al. (2025)'s figure even if more men than women use ChatGPT overall. [10] If women are more likely to use AI without an account (perhaps due to privacy concerns), this would also not be captured in their measure. In other words, parity among observed consumer accounts does not imply parity in the broader population.

More broadly, both the data from Chatterji et al. (2025) and our SimilarWeb data rely on imputed gender measures and don't measure the population of men and women at risk of using AI.[11] In contrast, the data in our systematic review both relies on more accurate self-reported gender measures and for 56 of the studies includes not just the gender breakdown of who is using AI, but the overall population at risk. Looking at Figure 2 reveals that even in early 2026, there appears to be a gap of roughly 5–13 percentage points across the studies we review.

While we have shown that there are significant gender gaps in who chooses to use AI, among visits

[10] For example, suppose the analyzed consumer-account sample contains 1,000 active accounts with typically feminine names and 1,000 with typically masculine names. If omitted corporate, education, enterprise, or logged-out use contains 250 women and 750 men, the broader active-user population would be only 41.7% female.

[11] Chatterji et al. (2025) classify gender only indirectly from first names, leaving a sizable "unknown" category. As their Figure 18 Panel A shows, roughly 20–30% of active accounts over much of the sample is "unknown" gender. Their estimated female share is higher when this unknown group is excluded from the denominator.

that do occur, do women and men engage with these tools for the same amount of time? If women are less likely to adopt generative AI and also spend less time using it, then overall adoption rates alone understate the AI gender gap. In Table 1, women's mean visit duration is significantly shorter on six of the top ten tools in the United States. On `chatgpt.com`, women spend 8.9% less time than men ($p < 0.001$), on `gemini.google.com`, 10.7% less ($p < 0.001$), on `grok.com`, 15.9% less ($p = 0.005$), on `perplexity.ai`, 13.9% less ($p < 0.001$), on `copilot.microsoft.com`, 4.9% less ($p < 0.001$), and on `chat.deepseek.com`, 8.9% less ($p = 0.004$). However, women do not spend less time than men on every tool. On `character.ai`, women's visits are 9.8% longer than men's ($p < 0.001$), and on `polybuzz.ai`, they are 11.6% longer ($p < 0.001$). Both of these tools are more conversational or companion-oriented platforms rather than general-purpose productivity tools.

Next, we again focus specifically on ChatGPT, which is by far the dominant product in our panel, accounting for 70.45% of all SimilarWeb panel traffic in the United States over the August 2025–January 2026 window (Table 1). In Figure 6, we plots age-weighted mean visit duration on `chatgpt.com` in the United States, separately by gender, from January 2023 through January 2026. This figure shows that women's average visit duration lies below men's in every month of the sample period. Women's visits are 10.4% shorter than men's on average, corresponding to about 33 fewer seconds per visit ($p < 2.2 \times 10^{-16}$). This gender gap intensity of use gap has also remained stable over time, with no consistent indication of narrowing as ChatGPT has become more popular and powerful.

This pattern that women use AI for less time than men is consistent with the broader literature on AI-use intensity from our systematic review. Several studies find that men use generative AI more frequently or persistently than women: Zahs, Schmodde, and Wehner (2026) finds that female employees use generative AI about 17.5 fewer minutes per day at work compared to their male colleagues; Kennedy et al. (2025) and Lean In (2026) report higher daily or near-daily AI use among men; and studies of clinicians, journalists, researchers, and UK adults similarly find male-favoring gaps in more frequent use (Egli et al., 2025; Thurman, Thäsler-Kordonouri, and Fletcher, 2025; Mohammadi et al., 2026; Stephany and Duszynski, 2026). Conditional on adoption, Gai, Hou, and Tu (2025) show that female software engineers send fewer prompts and copy fewer lines of AI-generated code, while Carvajal, Franco, and Isaksson (2024) finds that male students persist longer after an unsatisfactory AI response and write longer prompts. Across session duration, frequency, prompt volume, and persistence, the evidence shows that the gender gap extends beyond adoption to how intensively people use AI.

# 5 Evidence on Potential Mechanisms

Taken together, the systematic review and SimilarWeb observational evidence show a persistent gender gap in generative AI use across samples, settings, and intensity of use. We now turn to the mechanisms proposed in the existing literature that could plausibly generate these gaps. We organize this evidence around five frictions that are empirically and explicitly supported in our systematic review: knowledge and familiarity; perceived usefulness; institutional support, training, and confidence; social legitimacy and status costs; and trust, privacy, and risk concerns. Table 2 succinctly summarizes the mechanisms and the sources that support them.

These mechanisms differ in their implications as AI use becomes more mainstream and continues to diffuse. Knowledge, familiarity, and perceived usefulness may narrow over time as tools become more common, as more people observe concrete use cases and early uncertainty about returns declines. This interpretation is consistent with the descriptive evidence in Figure 2, where the gender gap has narrowed somewhat over time. However, the remaining mechanisms, such as institutional support, social legitimacy, and trust/risk concerns, are less likely to resolve through exposure alone and may explain why the gender gap has persisted. Because these mechanisms depend on workplace rules, managerial encouragement, norms, training access, social evaluation, risk perceptions, these point to the need for additional organizational, institutional, or tool change.

The most direct and most widely supported mechanism is knowledge and familiarity. Women report lower familiarity and self-assessed knowledge of generative AI in Aldasoro et al. (2024), Stöhr, Ou, and Malmström (2024), and Draxler et al. (2023). Female Swiss clinicians score lower on an objective LLM knowledge test in Egli et al. (2025). In their 2024 nationally representative survey of US household heads, Aldasoro et al. (2024) find that most of the gender gap they observe can be explained by women's lower self-assessed knowledge of generative AI, with self-rated knowledge explaining more than 70% of the reduction in the estimated gender gap after controls are added. In Chugunova et al. (2026), AI familiarity accounts for most of the gender AI gap among German researchers, with women more often citing lack of knowledge as a barrier. This mechanism is particularly important early in the diffusion process and for newer frontier tools. As public familiarity rises, the knowledge gap may shrink, increasing usage by different groups.

A second mechanism that may explain differences in AI use across genders is differences is perceived usefulness. Aldasoro et al. (2024) find that men see greater job-opportunity benefits from generative AI, while Carvajal, Franco, and Isaksson (2024) and Stöhr, Ou, and Malmström (2024) find that female students are less likely to view generative AI as useful for learning or coursework. In workplace surveys, Arbanas et al. (2024) report that female professionals are less likely than men to say AI currently boosts their productivity, and Lean In (2026) find that women are less likely to feel positive or energized about AI. Humlum and Vestergaard (2025) find no average gender gap in expected time savings from ChatGPT, but women are more uncertain about their beliefs about how much time they can save with AI. Surveying 400 U.S.-based academic LLM users in 2024, Tang et al. (2025) finds that men report greater efficiency gains from AI and have stronger intentions to recommend it to others. Zahs, Schmodde, and Wehner (2026) find that performance expectancy is the strongest predictor of AI usage intention, especially among women. The authors suggest that women may require stronger evidence of practical benefits before adopting AI, where men may be more willing to use it when performance gains are less certain. Like knowledge, perceived usefulness may increase with exposure. Once users observe concrete productivity gains, uncertainty about returns may fall.

The next three mechanisms are especially relevant for explaining persistence in the gap. A third mechanism is institutional support, training, and confidence. Arbanas et al. (2024) find that female users are less likely than male users to say their employer actively encourages AI use or invests in training. Lean In (2026) similarly finds that men are more likely than women to say their manager encourages AI use. Barisano et al. (2024) find that junior women are less willing than junior men to use AI in the absence of clear company policy, Chugunova et al. (2026) find that perceived institutional regulations are more negatively associated with women's AI use, and Carvajal, Franco, and Isaksson (2024) and Contractor and Reyes (2025) find that female students respond more strongly to explicit bans than their male peers. Humlum and Vestergaard (2025) similarly find that women are more likely to say they do not know how to use ChatGPT and that they report needing training. Using a survey experiment, Carvajal, Franco, and Isaksson (2024) finds that women report not only lower confidence in prompting AI, but have lower persistence when prompts fail, and have lower average prompt success. Chugunova et al. (2026) found that women are less likely to use or intend to use prompt-engineering resources. Monday Girl (2026) studying Canadians in February 2026 found that women report lower confidence and comfort using AI and higher intimidation and confusion compared to men. A pilot in the UK conducted by (Public First and Google, 2025) finds that after an AI training, women's daily and weekly AI use rising sharply, and participants most commonly selected "I understand more about how AI tools work" as the reason their usage increased. (Public First and Google, 2025). This mechanism helps explain why gender gaps vary so much across settings—the same individual may behave differently depending on whether AI use is encouraged, prohibited, ambiguous, or accompanied by training for a specific context.

A fourth mechanism is social legitimacy and status costs. In student settings, Carvajal, Franco, and Isaksson (2024) find that women are more likely to view ChatGPT use as cheating, and Stöhr, Ou, and Malmström (2024) find that female students are more likely to say AI use goes against the purpose of education and should be prohibited. Stone (2025) similarly finds that men report greater moral acceptability of AI use in banned or ambiguous academic contexts. In workplace settings, Lean In (2026) find that women are more likely to worry AI use will be perceived as cheating and less likely to report being praised for using it. The clearest evidence on status costs comes from Gai, Hou, and Tu (2025), who show that identical AI-assisted work receives a competence penalty, especially when the engineer is female, and that female engineers anticipate larger penalties than male engineers. Similarly, Monday Girl (2026) find that women are more likely than men to worry about negative workplace consequences from AI use.

A fifth mechanism relates to differences in trust, privacy, and risk concerns. Arbanas et al. (2024) find that women report lower trust that AI providers will keep their data secure and stronger privacy concerns. In Aldasoro et al. (2024), privacy and trust concerns explain part of the gender gap. Stephany and Duszynski (2026) find that concerns about mental health, privacy, climate, and labor-market effects depress adoption more strongly for women than men in the UK, and Borwein et al. (2026) studying US and Canadian adults in 2023, find that women are 11% more likely than men to say AI's risks exceed its benefits. Concerns about quality, accuracy, personal style, creativity, and ethical risks appear in several sources (Draxler et al., 2023; Egli et al., 2025). Bachaud (2026) surveying job seekers in 2026, find that women are more likely to be very concerned about AI-led dehumanization in hiring. The gender-specific evidence on trust in AI accuracy is mixed. Lean In (2026) find that women are more likely to question AI accuracy, while Carvajal, Franco, and Isaksson (2024), Barisano et al. (2024), and Chugunova et al. (2026) find little or no evidence that women are generally less trusting of AI accuracy or returns. Personal job-loss fears because of AI do not appear to be a primary driver of the adoption gaps, since these concerns are rare for women in Humlum and Vestergaard (2025) and Howington (2024) finds greater concern among men. However, Lean In (2026) find that women are much more likely to expect other women to be laid off because of AI, and Borwein et al. (2026) links women's AI skepticism to greater perceived and actual exposure to AI-driven labor-market risk.

These latter mechanisms (institutional support, social legitimacy, and trust, privacy, and risk concerns) are where organizational choices matter most. Policies that clarify acceptable use, manager encouragement, training access, privacy protections, and explicit norms around AI-assisted work can meaningfully change adoption incentives without requiring differences in underlying ability or knowledge. They are also the mechanisms our SimilarWeb data is least equipped to identify directly—traffic data can show persistent differences in use, intensity, and platform choice, but we are unable to see whether a user avoided AI because of workplace rules, fear of competence penalties, privacy concerns, poor tool design, or other frictions. Further research is needed on the institutional and design-side mechanisms behind persistence: whether organizations provide clear policies and training, whether users face social or reputational penalties for AI-assisted work (or just perceive that they do), and the extent to which tool design, protections, and organizational choices can make this new technology feel trustworthy, learnable, and useful to different groups.

## 6 Conclusion

Our findings document that gender gaps in generative AI are nearly universal and persistent. We find that women use generative AI less than men in a systematic review covering 76 sources as well as in web analytics data on who visits the leading generative AI websites worldwide. The gap has narrowed over the last three years of rapid diffusion but appears to have stabilized. The gap is present within tightly bounded occupations and single institutions, indicating that it is not reducible to occupational segregation. We also see gaps on both the extensive and the intensive margin, finding that conditional on adoption, women send fewer prompts, log fewer daily minutes of use, and in many tools have significantly shorter visits than men. Across the 10 most visited AI tools, the share of traffic from female users in the United States ranges from 44.1% on `chatgpt.com`, down to 26.7% on `grok.com`. Overall, the gender gaps in AI adoption and use seem both widespread and persistent.

These findings appear to be driven by two sets of mechanisms with differing implications for how the gender gap in AI usage will evolve. First, there are explanations rooted in knowledge and familiarity. These explanations are ones the market should close over time. That said, we want to caution that if diffusion takes years to close the gap, this can still have meaningful consequences for careers, as early adopters accumulate value experience in learning how to work with tools and capture first mover advantages (Lieberman and Montgomery, 1988).

The second set of mechanisms is rooted in institutional, organizational, and social differences across genders. This is consistent with a long line of work showing that the implications of new workplace technologies are produced through interaction between the tool and the social structures of roles, routines, and evaluation (Orlikowski, 1992; Barley, 1986). These gaps can be produced or mitigated by organizational choices, including how firms introduce AI, what training looks like, who is praised for using it, and how AI-assisted work is evaluated.

These gaps could have real consequences for women and their careers. As generative AI systems are still in their formative stages, the under-representation of women may result in biases in the user data these tools learn from, leading to self-reinforcing gender disparities (Cao, Koning, and Nanda, 2024). Such biases in user data could result in generative AI systems that reinforce gendered stereotypes and in tools that are less effective at the tasks more often performed by women (Koenecke et al., 2020; Guilbeault et al., 2024). Ensuring that generative AI tools are designed inclusively and do not inadvertently substitute for broader efforts to reduce gendered inequalities will be crucial for unlocking their full potential to enhance productivity and to reduce inequality (Pugh, 2024). That said, the impact of AI on careers is far from certain. Paradoxically, it is possible that the same lower adoption that disadvantages women when early use is rewarded could potentially benefit them if AI reduces learning, erodes skills, or weakens critical thinking. In fact, recent evidence has shown that AI use has increased the quantity but reduced the quality of academic writing in the social sciences (Gartenberg et al., 2026). Overall, it is possible that the gender gaps in AI in our study have broader implications for gender gaps in society.

While our study provides substantial systematic evidence showing gender gaps in who uses AI, it is not without limitations. First, we use aggregated data from our systematic review (which primarily consists of surveys) and website traffic data from SimilarWeb, which makes it difficult for us to compare differences across samples and studies across countries or over time or track adoption by individuals longitudinally. While these limitations do not invalidate our estimate of the gender gap in usage, they do limit our ability to pin down more precise gaps in subgroups and identify the underlying mechanisms. That said, while systematic reviews are limited in their ability to pin down mechanisms, their primary advantage lies in providing a more complete map of a topic, guarding against over-indexing on a single result. In this case, our paper shows that the gender gap has yet to close. While the gap is narrowing and in some samples has slightly flipped (Chatterji et al. (2025)), on the whole, our evidence points to meaningful and persistent gaps that research must address.

These gender gaps are meaningful. Given estimates that generative AI has the potential to increase US economic output and worker productivity levels by nearly 20% over the next decade (Baily, Brynjolfsson, and Korinek, 2023), because women make up just under 50% of the US workforce, a persistent 16% usage gap could result in hundreds-of-billions of dollars of lost productivity and growth in the US alone. Specifically, if women's lower usage persists, we risk delaying, or missing out on completely, the work, inventions, and ideas women would have produced with this new technology (Koning, Samila, and Ferguson, 2021). Mapping where these gaps are largest, which tools they affect, and how they appear in adoption and intensity of use is therefore essential for helping scholars, managers, and policymakers target scarce time and resources. Without intervention, generative AI's potential to drive economic growth and improve productivity will almost surely disproportionately benefit male users, with the potential to further entrench existing gender gaps.

## References


Ahmed, Weam Mohamed Meargni, Malaz M. Abdalmotalib, Mohamed H. Elbadawi, Galia Tajelsir Fadulelmula Mohammed, Waad Mohamed Ibrahim Mohamed, Fatima Salih Babiker Mohammed, Hajar Saad Salih, and Hiba Omer Yousif Mohamed. 2025. "Shaping the Future of Medical Education: A Cross-Sectional Study on ChatGPT Attitude and Usage Among Medical Students in Sudan." *PLOS ONE* 20 (5):e0324698.

Alammari, Dalia, Elaf Alamari, Rawan Alamri, Raneem Alharbi, Jumana Felimban, and Jana Aljohani. 2025. "Assessing Medical Students' Attitudes, Performance, and Usage of ChatGPT in Jeddah, Saudi Arabia." *Frontiers in Artificial Intelligence* 8:1577911.

Aldasoro, Iñaki, Olivier Armantier, Sebastian Doerr, Leonardo Gambacorta, and Tommaso Oliviero. 2024. "The gen AI gender gap." *Economics Letters* 241:111814.

Alonso-Muñoz, Laura and Andreu Casero-Ripollés. 2026. "Adoption of Generative AI in Higher Education: Perceptions of Journalism Students." *Information* 17 (2):189.

AmeriSpeak and NORC at the University of Chicago. 2025. "AI Adoption Report: Tracking the Rise of AI in Americans' Lives." Tech. rep., AmeriSpeak Omnibus / NORC at the University of Chicago. Q3 2025 report; August 22–24, 2025 wave used in the meta-analysis.

Angrisani, Marco, Maria Casanova, Nathanael J. Fast, Jimmy Narang, and Juliana Schroeder. 2026. "Gaps in large language model awareness, usage, and perceptions in the United States: Evidence from a nationally representative longitudinal survey." *PNAS Nexus* 5 (2):pgag007.

Arbanas, Jana, Paul H. Silverglate, Susanne Hupfer, Jeff Loucks, Prashant Raman, and Michael Steinhart. 2024. "Women and generative AI: The adoption gap is closing fast, but a trust gap persists." Tech. rep., Deloitte Insights, TMT Predictions 2025. URL https://www.deloitte.com/us/en/insights/industry/technology/technology-media-and-telecom-predictions/2025/women-and-generative-ai.html.

Arntz, Melanie, Myriam Baum, Eduard Brüll, Ralf Dorau, Matthias Hartwig, Britta Matthes, Sophie-Charlotte Meyer, Oliver Schlenker, Anita Tisch, and Sascha Wischniewski. 2026. "Low Barriers, High Stakes: Formal and Informal Diffusion of AI in the Workplace." Tech. Rep. 26-001, ZEW Discussion Paper. Also circulated as IAB-Discussion Paper 422.

Autor, David. 2024. "Applying AI to rebuild middle class jobs." Tech. rep., National Bureau of Economic Research.

Bachaud, Nicole. 2026. "AI-Powered Job Seekers Are Twice as Likely to Land an Offer." ZipRecruiter Economic Research. URL https://www.ziprecruiter-research.org/commentary/ai-powered-job-seekers. Published 30 April 2026; Q1 2026 Job Seeker Confidence Survey.

Baily, Martin Neil, Erik Brynjolfsson, and Anton Korinek. 2023. "Machines of mind: How generative AI will power the coming productivity boom." *Brookings Institution* URL https://www.brookings.edu/articles/machines-of-mind-how-generative-ai-will-power-the-coming-productivity-boom/. Accessed: 2024-10-11.

Barisano, Maria, Neveen Awad, Adriana Dahik, Julie Bedard, Uche Monu, Gunjan Mundhra, and Katherine Lou. 2024. "Women Leaders in Tech Are Paving the Way in GenAI." *BCG* URL https://www.bcg.com/publications/2024/women-leaders-in-tech-are-paving-the-way-in-genai. Accessed: 2024-07-16.

Barley, Stephen R. 1986. "Technology as an occasion for structuring: Evidence from observations of CT scanners and the social order of radiology departments." *Administrative science quarterly* :78–108.

Baron, James N and Jeffrey Pfeffer. 1994. "The social psychology of organizations and inequality." *Social Psychology Quarterly* :190–209.

Bianchi, Suzanne M, Liana C Sayer, Melissa A Milkie, and John P Robinson. 2012. "Housework: Who did, does or will do it, and how much does it matter?" *Social forces* 91 (1):55–63.

Bick, Alexander, Adam Blandin, and David J Deming. 2024. "The Rapid Adoption of Generative AI." *Working paper* .

Bick, Alexander, Adam Blandin, David J. Deming, Nicola Fuchs-Schündeln, and Jonas Jessen. 2026. "Mind the Gap: AI Adoption in Europe and the US." Tech. Rep. 34995, National Bureau of Economic Research. Also presented at Brookings Papers on Economic Activity, Spring 2026.

Biernat, Monica and Melvin Manis. 1994. "Shifting standards and stereotype-based judgments." *Journal of personality and social psychology* 66 (1):5.

Bimber, Bruce. 2000. "Measuring the Gender Gap on the Internet." *Social Science Quarterly* 81 (3):868–876. URL https://www.jstor.org/stable/42864010.

Björkegren, Daniel. 2023. "Artificial Intelligence for the Poor: How to Harness the Power of AI in the Developing World." Foreign Affairs.

Björkegren, Daniel, Jun Ho Choi, Divya Panchaksharappa Budihal, Dominic Sobhani, Oliver Garrod, and Paul Atherton. 2025. "Could AI Leapfrog the Web? Evidence from Teachers in Sierra Leone." *Working paper, Columbia University* .

Borwein, Sophie, Beatrice Magistro, R. Michael Alvarez, Bart Bonikowski, and Peter J. Loewen. 2026. "Explaining women's skepticism toward artificial intelligence: The role of risk orientation and risk exposure." *PNAS Nexus* 5 (1):pgaf399.

Brooks, Chad. 2023. "ChatGPT Usage Rates Among American Workers: Study." Business.com. URL https://www.business.com/articles/chatgpt-usage-rates-among-american-workers-study/. Paper included in meta-analysis corpus.

Brynjolfsson, Erik, Danielle Li, and Lindsey R. Raymond. 2025. "Generative AI at Work." *The Quarterly Journal of Economics* 140 (2):889–942.

Campus AI. 2025. "Generative AI Adoption: The Report, Vol. 1." Tech. rep., Campus AI.

Cao, Ruiqing, Rembrand Koning, and Ramana Nanda. 2024. "Sampling Bias in Entrepreneurial Experiments." *Management Science* 70 (10):7283–7307.

Carvajal, Daniel, Catalina Franco, and Siri Isaksson. 2024. "Will Artificial Intelligence Get in the Way of Achieving Gender Equality?" *NHH Dept. of Economics Discussion Paper* (03).

Castilla, Emilio J. 2008. “Gender, Race, and Meritocracy in Organizational Careers.” *American Journal of Sociology* 113 (6):1479–1526.

Cech, Erin, Brian Rubineau, Susan Silbey, and Carroll Seron. 2011. “Professional Role Confidence and Gendered Persistence in Engineering.” *American Sociological Review* 76 (5):641–666.

Chatterji, Aaron, Thomas Cunningham, David J. Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. 2025. “How People Use ChatGPT.” Tech. Rep. 34255, National Bureau of Economic Research.

Chugunova, Marina, Dietmar Harhoff, Katharina Hölzle, Verena Kaschub, Sonal Malagimani, Ulrike Morgalla, and Robert Rose. 2026. “Who uses AI in research, and for what? Large-scale survey evidence from Germany.” *Research Policy* 55 (2):105381.

Coffman, Katherine Baldiga. 2014. “Evidence on Self-Stereotyping and the Contribution of Ideas.” *The Quarterly Journal of Economics* 129 (4):1625–1660.

Conceição, Pedro et al. 2025. “A matter of choice: People and possibilities in the age of AI.” Tech. rep., United Nations Development Programme (UNDP) Human Development Report 2025.

Contractor, Zara and Germán Reyes. 2025. “Generative AI in Higher Education: Evidence from an Elite College.” *arXiv:2508.00717* .

Correll, Shelley J. 2001. “Gender and the career choice process: The role of biased self-assessments.” *American journal of Sociology* 106 (6):1691–1730.

Correll, Shelley J. 2004. “Constraints into Preferences: Gender, Status, and Emerging Career Aspirations.” *American Sociological Review* 69 (1):93–113.

Correll, Shelley J and Cecilia L Ridgeway. 2003. “Expectation states theory.” In *Handbook of social psychology*. Springer, 29–51.

Croson, Rachel and Uri Gneezy. 2009. “Gender differences in preferences.” *Journal of Economic literature* 47 (2):448–474.

Delecourt, Solène and Anne Fitzpatrick. 2021. “Childcare matters: Female business owners and the baby-profit gap.” *Management Science* 67 (7):4455–4474.

Dell’Acqua, Fabrizio, Edward McFowland, Ethan R Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Krayer, François Candelon, and Karim R Lakhani. 2023. “Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality.” *Working Paper available at SSRN 4573321* .

DiMaggio, Paul and Filiz Garip. 2012. “Network effects and social inequality.” *Annual review of sociology* 38 (1):93–118.

Dorta-González, Pablo, Alexis Jorge López-Puig, María Isabel Dorta-González, and Sara M. González-Betancor. 2024. “Generative artificial intelligence usage by researchers at work: Effects of gender, career stage, type of workplace, and perceived barriers.” *Telematics and Informatics* 94:102187.

Draxler, Fiona, Daniel Buschek, Mikke Tavast, Perttu Hämäläinen, Albrecht Schmidt, Juhi Kulshrestha, and Robin Welsch. 2023. “Gender, Age, and Technology Education Influence the Adoption and Appropriation of LLMs.” *arXiv preprint arXiv:2310.06556* .

Egli, Simon Bruno, Armon Arpagaus, Simon Adrian Amacher, Sabina Hunziker, and Stefano Bassetti. 2025. "Use, knowledge and perception of large language models in clinical practice: a cross-sectional mixed-methods survey among clinicians in Switzerland." *BMJ Health & Care Informatics* 32:e101470.

El-Sobkey, Salwa B., Mahmoud ElKholy, Mariam Abdallah, Dina Al-Amir Mohamed, and Tayseer Abdeldayem. 2026. "Use of ChatGPT in Physical Therapy Education: Comparative Study of Usefulness to Students and Ease of Use Perceptions, and Descriptions of Usage Patterns." *Journal of Taibah University Medical Sciences* 21 (1):178–189.

England, Paula. 2010. "The Gender Revolution: Uneven and Stalled." *Gender & Society* 24 (2):149–166.

England, Paula, Andrew Levine, and Emma Mishel. 2020a. "Progress toward gender equality in the United States has slowed or stalled." *Proceedings of the National Academy of Sciences* 117 (13):6990–6997.

———. 2020b. "Progress toward Gender Equality in the United States Has Slowed or Stalled." *Proceedings of the National Academy of Sciences* 117 (13):6990–6997.

English, Patrick et al. 2026. "Public adoption, sentiment, feelings, and regulatory demands on artificial intelligence, big tech, and social media in Western societies." YouGov. URL https://yougov.co.uk/technology/articles/53925-public-adoption-sentiment-feelings-and-regulatory-demands-on-ai-big-tech-a Paper included in meta-analysis corpus.

Eurostat. 2026. "Use of Artificial Intelligence by Individuals." Statistics Explained. URL https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_by_individuals. Data extracted March 2026; article source dated 15 April 2026.

Exley, Christine L. and Judd B. Kessler. 2022. "The Gender Gap in Self-Promotion." *The Quarterly Journal of Economics* 137 (3):1345–1381.

Gai, Phyliss Jia, Jiayi Hou, and Yanping Tu. 2025. "Competence Penalty Is a Barrier to the Adoption of New Technology." *Working paper, Peking University* .

Gambacorta, Leonardo, Tullio Jappelli, and Tommaso Oliviero. 2025. "Exploring household adoption and usage of generative AI: new evidence from Italy." Tech. Rep. 1298, BIS Working Papers, Bank for International Settlements.

Gartenberg, Claudine, Sharique Hasan, Alex Murray, and Lamar Pierce. 2026. "More Versus Better: Artificial Intelligence, Incentives, and the Emerging Crisis in Peer Review." *Organization Science* 37 (3). URL https://pubsonline.informs.org/doi/10.1287/orsc.2026.ed.v37.n3. Published online April 27, 2026.

Glassdoor Economic Research. 2024. "Conversation Starter: ChatGPT Usage in the Workplace More than Doubles Year Over Year." *Glassdoor* .

Gnambs, Timo, Jan-Philipp Stein, Sabine Zinn, Florian Griese, and Markus Appel. 2025. "Attitudes, experiences, and usage intentions of artificial intelligence: A population study in Germany." *Telematics and Informatics* 98:102265.

Goldin, Claudia. 2014. "A grand gender convergence: Its last chapter." *American economic review* 104 (4):1091–1119.

———. 2021. *Career and family: Women's century-long journey toward equity*. Princeton University Press.

Goligoski, Emily, Jacob Clemente, Chris Miller, Kevin Delaney, and Daniel Lee. 2024. "Playbook: Using AI in Ways That Enhance Worker Dignity and Inclusion." *Charterworks* .

Gonzalez Vazquez, Ignacio, Enrique Fernandez Macias, Sally Wright, and Davide Villani. 2025. "Digital Monitoring, Algorithmic Management and the Platformisation of Work in Europe." Tech. rep., Joint Research Centre, European Commission.

Google and Ipsos. 2024. "Our life with AI: The reality of today and the promise of tomorrow." Tech. rep., Google and Ipsos.

Guilbeault, Douglas, Solène Delecourt, and Bhargav Srinivasa Desikan. 2025. "Age and gender distortion in online media and large language models." *Nature* 646 (8087):1129–1137.

Guilbeault, Douglas, Solène Delecourt, Tasker Hull, Bhargav Srinivasa Desikan, Mark Chu, and Ethan Nadler. 2024. "Online images amplify gender bias." *Nature* 626 (8001):1049–1055.

Hartley, Jonathan S., Filip Jolevski, Vitor Melo, and Brendan Moore. 2025. "The Labor Market Effects of Generative Artificial Intelligence." *SSRN working paper* URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5136877. Previously cited as 2026; public survey data available at https://generativeairesearch.org/us-survey-data/.

Haslberger, Matthias, Jane Gingrich, and Jasmine Bhatia. 2023. "No Great Equalizer: Experimental Evidence on AI in the UK Labor Market." *Working paper, University of St. Gallen* .

Howington, Jessica. 2024. "The AI Gender Gap: Exploring Variances in Workplace Adoption." *FlexJobs Blog* URL https://www.flexjobs.com/blog/post/the-ai-gender-gap-exploring-variances-in-workplace-adoption/. Accessed: 2024-07-16.

Humlum, Anders and Emilie Vestergaard. 2025. "The unequal adoption of ChatGPT exacerbates existing inequalities among workers." *Proceedings of the National Academy of Sciences* 122 (1):e2414972121.

Ibrahim, Fabio, Johann-Christoph Münscher, Monika Daseking, and Nils-Torge Telle. 2025. "The technology acceptance model and adopter type analysis in the context of artificial intelligence." *Frontiers in Artificial Intelligence* .

Kennedy, Brian, Alec Tyson, Cary Funk et al. 2025. "How the US Public and AI Experts View Artificial Intelligence." Pew Research Center. URL https://www.pewresearch.org/internet/2025/04/03/how-the-us-public-and-ai-experts-view-artificial-intelligence/.

Kim, Hyunjin, Dahyeon Kim, and Rembrand Koning. 2026. "Mapping AI into Production: A Field Experiment on Firm Performance." .

Kirk, Hannah Rose, Yennie Jun, Filippo Volpin, Haider Iqbal, Elias Benussi, Frederic Dreyer, Aleksandar Shtedritski, and Yuki Asano. 2021. "Bias out-of-the-box: An empirical analysis of intersectional occupational biases in popular generative language models." *Advances in Neural Information Processing Systems* 34:2611–2624.

Koenecke, Allison, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R Rickford, Dan Jurafsky, and Sharad Goel. 2020. "Racial disparities in automated speech recognition." *Proceedings of the National Academy of Sciences* 117 (14):7684–7689.

Koning, Rembrand, Sharique Hasan, and Aaron Chatterji. 2022. "Experimentation and start-up performance: Evidence from A/B testing." *Management Science* 68 (9):6434–6453.

Koning, Rembrand, Sampsa Samila, and John-Paul Ferguson. 2021. "Who do we invent for? Patents by women focus more on women's health, but few women get to invent." *Science* 372 (6548):1345–1348.

Kreacic, Ana and Terry Stone. 2024. "Women are Falling Behind on Generative AI in the Workplace. Here's How to Change That." *World Economic Forum* URL https://www.weforum.org/agenda/2024/04/women-generative-ai-workplace/. Accessed: 2024-07-16.

Lean In. 2026. "Women and AI: The Gender Gap in AI Adoption and Recognition." Lean In Research. URL https://leanin.org/research/ai-women-gender-gap-data.

Lieberman, Marvin B. and David B. Montgomery. 1988. "First-Mover Advantages." *Strategic Management Journal* 9 (S1):41–58.

Liu, Yan and He Wang. 2026. "Who on earth is using Generative AI?" *World Development* 199:107260.

Lopez Barrios, Michel and Catherine E. Déri. 2025. "The Usage of Generative Artificial Intelligence According to Sociodemographic Characteristics of Undergraduate Students." *Journal of Scholarly Publishing* .

Loschiavo, David, Olivier Armantier, Antonio Dalla Zuanna, Leonardo Gambacorta, Mirko Moscatelli, and Ilaria Supino. 2026. "Embracing gen AI: a comparison of Italian and US households." Tech. Rep. 1322, BIS Working Papers, Bank for International Settlements.

Lyttelton, Thomas, Maxim Massenkoff, and Nathan Wilmers. 2026. "Coding Agents in the Social Sciences." URL https://www.anthropic.com/research/coding-agents-social-sciences.

Maaß, L., C. Grab-Kroll, J. Koerner, W. Öchsner, M. Schön, D. A. C. Messerer, T. M. Böckers, and Anja Böckers. 2025. "Artificial Intelligence and ChatGPT in Medical Education: A Cross-Sectional Questionnaire on Students' Competence." *Journal of CME* 14:2437293.

Maslej, Nestor, Loredana Fattorini, Raymond Perrault, Vanessa Parli, Anka Reuel, Erik Brynjolfsson, John Etchemendy, Katrina Ligett, Terah Lyons, James Manyika et al. 2024. "Artificial intelligence index report 2024." *arXiv preprint arXiv:2405.19522* .

Merton, Robert K. 1968. "The Matthew Effect in Science." *Science* 159 (3810):56–63.

Møgelvang, Anja, Camilla Bjelland, Simone Grassini, and Kristine Ludvigsen. 2024. "Gender Differences in the Use of Generative Artificial Intelligence Chatbots in Higher Education: Characteristics and Consequences." *Education Sciences* 14 (12):1363.

Mohammadi, Ehsan, Mike Thelwall, Yizhou Cai, Taylor Collier, Iman Tahamtan, and Azar Eftekhar. 2026. "Is generative AI reshaping academic practices worldwide? A survey of adoption, benefits, and concerns." *Information Processing & Management* 63 (1):104350.

Monday Girl. 2026. "Nearly Half of Women Concerned About Backlash From AI Use At Work, New Data Finds." Business Wire. URL https://www.businesswire.com/news/home/20260310249264/en/. Published 10 March 2026; survey fielded February 24–26, 2026 among Angus Reid Forum adult Canadians.

Nam, Jane. 2024. "56% of College Students Have Used AI on Assignments or Exams." *BestColleges* URL https://www.bestcolleges.com/research/most-college-students-have-used-ai-survey/. Accessed: 2024-07-16.

Nordling, Linda. 2023. "How ChatGPT is transforming the postdoc experience." *Nature* 622 (7983):655–657.

Notley, Tanya, Simon Chambers, Sora Park, and Michael Dezuanni. 2024. "Adult Media Literacy in 2024: Australian Attitudes, Experiences and Needs." Tech. rep., Western Sydney University and Queensland University of Technology.

Noy, Shaked and Whitney Zhang. 2023. "Experimental evidence on the productivity effects of generative artificial intelligence." *Science* 381:187–192.

Ofosu-Ampong, Kingsley. 2023. "Gender, Digital Technology and Artificial Intelligence: Perspectives on Adoption and Use." *Journal of Digital Art & Humanities* 4 (2).

Orlikowski, Wanda J. 1992. "The duality of technology: Rethinking the concept of technology in organizations." *Organization science* 3 (3):398–427.

Otis, Nicholas, Rowan P Clarke, Solène Delecourt, David Holtz, and Rembrand Koning. 2023. "The uneven impact of generative AI on entrepreneurial performance." *Working paper available at SSRN 4671369* .

Park, Eugenie and Risa Gelles-Watnick. 2023. "Most Americans haven't used ChatGPT; few think it will have a major impact on their job." *Pew Research Center* 28.

Pérez-Portabella, Antonio, Jorge de Andrés-Sánchez, Mario Arias-Oliva, and Mar Souto-Romero. 2025. "Integrating Machine Learning Techniques and the Theory of Planned Behavior to Assess the Drivers of and Barriers to the Use of Generative Artificial Intelligence: Evidence in Spain." *Algorithms* 18 (7):410.

Petersen, Trond and Laurie A. Morgan. 1995. "Separate and Unequal: Occupation-Establishment Sex Segregation and the Gender Wage Gap." *American Journal of Sociology* 101 (2):329–365.

Pizarro Milian, Roger and Julie Peters. 2024. "Generative AI Use: Taking Stock of Gender Disparities Over Time." Academica Forum. URL https://forum.academica.ca/forum/generative-ai-use-taking-stock-of-gender-disparities-over-time. Published 23 October 2024.

Public First and Google. 2025. "AI Works: A people-first skills pilot, exploring AI adoption in the workplace." Research report, Public First and Google. Fieldwork August 2024 — March 2025; second landscape poll fielded February 28 — March 7, 2025.

Pugh, Allison J. 2024. *The Last Human Job: The Work of Connecting in a Disconnected World.* Princeton University Press.

Ravšelj, Dejan, Damijana Keržič, Nina Tomaževič et al. 2025. "Higher education students' perceptions of ChatGPT: A global study of early reactions." *PLOS ONE* 20 (2):e0315011.

Reskin, Barbara. 1993. "Sex segregation in the workplace." *Annual review of sociology* 19 (1):241–270.

Ridgeway, Cecilia L. 2011. *Framed by gender: How gender inequality persists in the modern world.* Oxford University Press.

Ridgeway, Cecilia L and Shelley J Correll. 2004. "Unpacking the gender system: A theoretical perspective on gender beliefs and social relations." *Gender & society* 18 (4):510–531.

Rogers, Everett M, Arvind Singhal, and Margaret M Quinlan. 2019. "Diffusion of innovations." In *An integrated approach to communication theory and research*. Routledge, 415–434.

Russo, Claudia, Luca Romano, Davide Clemente, Leonardo Iacovone, Thomas E. Gladwin, and Angelo Panno. 2025. "Gender differences in artificial intelligence: the role of artificial intelligence anxiety." *Frontiers in Psychology* 16:1559457.

Saad, Lydia. 2025. "AI: Major Threat or Just the Next Tech Thing?" Gallup. URL https://news.gallup.com/poll/692435/major-threat-next-tech-thing.aspx.

Salesforce, Slack, and YouGov. 2024. "Australians Embracing Generative AI at Work, but Trust Gap Remains." Salesforce News & Insights ANZ. URL https://www.salesforce.com/au/news/stories/ai-work-trust/. Published 27 February 2024.

Sallam, Malik, Nesreen A. Salim, Muna Barakat, Kholoud Al-Mahzoum, Ala'a B. Al-Tammemi, Diana Malaeb, Rabih Hallit, and Souheil Hallit. 2023. "Assessing Health Students' Attitudes and Usage of ChatGPT in Jordan: Validation Study." *JMIR Medical Education* 9:e48254.

Sarasohn, Eliza. 2024. "Despite AI Enthusiasm, Workforce Index Reveals Workers Aren't Yet Unlocking Its Benefits." Slack Blog. URL https://slack.com/blog/news/the-workforce-index-june-2024. Paper included in meta-analysis corpus.

Sarsons, Heather. 2017. "Recognition for group work: Gender differences in academia." *American Economic Review* 107 (5):141–145.

Shishakly, Rima. 2025. "Understanding AI in Higher Education: Gendered and Intersectional Students' Experiences with ChatGPT Use." *European Journal of STEM Education* 10 (1):36.

Stephany, Fabian and Jedrzej Duszynski. 2026. "Women Worry, Men Adopt: How Gendered Perceptions Shape the Use of Generative AI." *arXiv preprint arXiv:2601.03880* .

Stöhr, Christian, Amy Wanyu Ou, and Hans Malmström. 2024. "Perceptions and usage of AI chatbots among students in higher education across genders, academic levels and fields of study." *Computers and Education: Artificial Intelligence* :100259.

Stone, Brian W. 2025. "Generative AI in Higher Education: Uncertain Students, Ambiguous Use Cases, and Mercenary Perspectives." *Teaching of Psychology* .

Sublime, Jérémie and Ilaria Renna. 2024. "Is ChatGPT Massively Used by Students Nowadays? A Survey on the Use of Large Language Models such as ChatGPT in Educational Settings." *arXiv preprint arXiv:2412.17486* URL https://arxiv.org/abs/2412.17486.

Suh, Donghyun and Samil Oh. 2026. "Generative AI and the Reallocation of Time: Productivity, Leisure, and Fulfilling Work." *arXiv:2602.12695* .

Tang, Chuang, Shaobo (Kevin) Li, Suming Hu, Fue Zeng, and Qianzhou Du. 2025. "Gender disparities in the impact of generative artificial intelligence: Evidence from academia." *PNAS Nexus* 4 (2):pgae591.

Thurman, Neil, Sina Thäsler-Kordonouri, and Richard Fletcher. 2025. "AI Adoption by UK Journalists and their Newsrooms: Surveying Applications, Approaches, and Attitudes." Tech. rep., Reuters Institute for the Study of Journalism, Oxford.

Tomaskovic-Devey, Don and Sheryl Skaggs. 2002. "Sex segregation, labor process organization, and gender earnings inequality." *American Journal of Sociology* 108 (1):102–128.

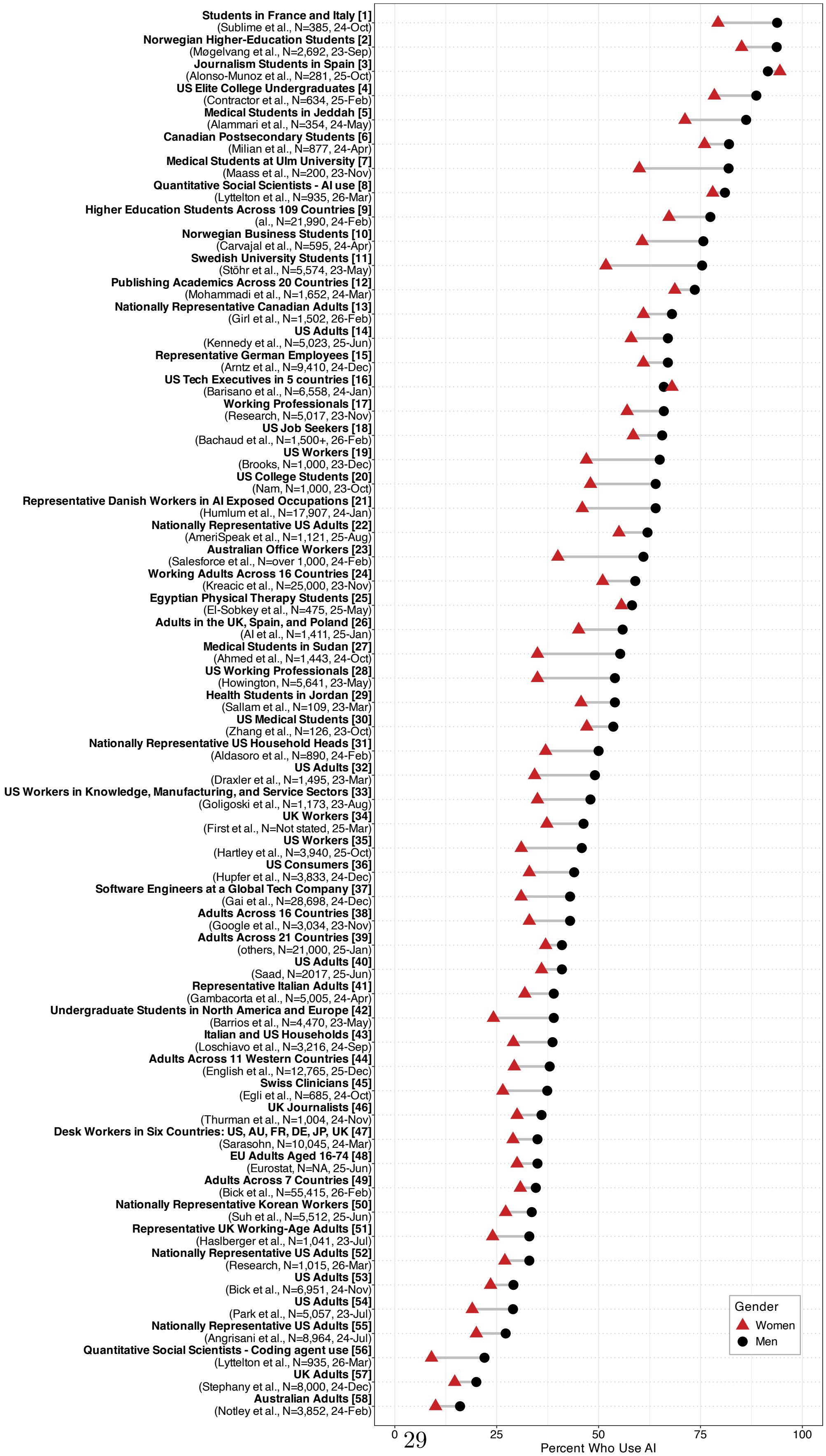
Van Noorden, Richard and Jeffrey M Perkel. 2023. "AI and science: what 1,600 researchers think." *Nature* 621 (7980):672–675.

Wiles, Emma, Lisa Krayer, Mohamed Abbadi, Urvi Awasthi, Ryan Kennedy, Pamela Mishkin, Daniel Sack, and François Candelon. 2026. "Generative AI and the Temporary Upskilling of Knowledge Workers." *Nature Human Behaviour* URL https://emmawiles.github.io/storage/reskill.pdf. Forthcoming. Previously titled "Generative AI as an Exoskeleton".

Zahs, Dominik, Lynn Schmodde, and Marius C. Wehner. 2026. "Mind the Gap: Gender Differences in Generative AI Adoption at Work." In *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS)*.

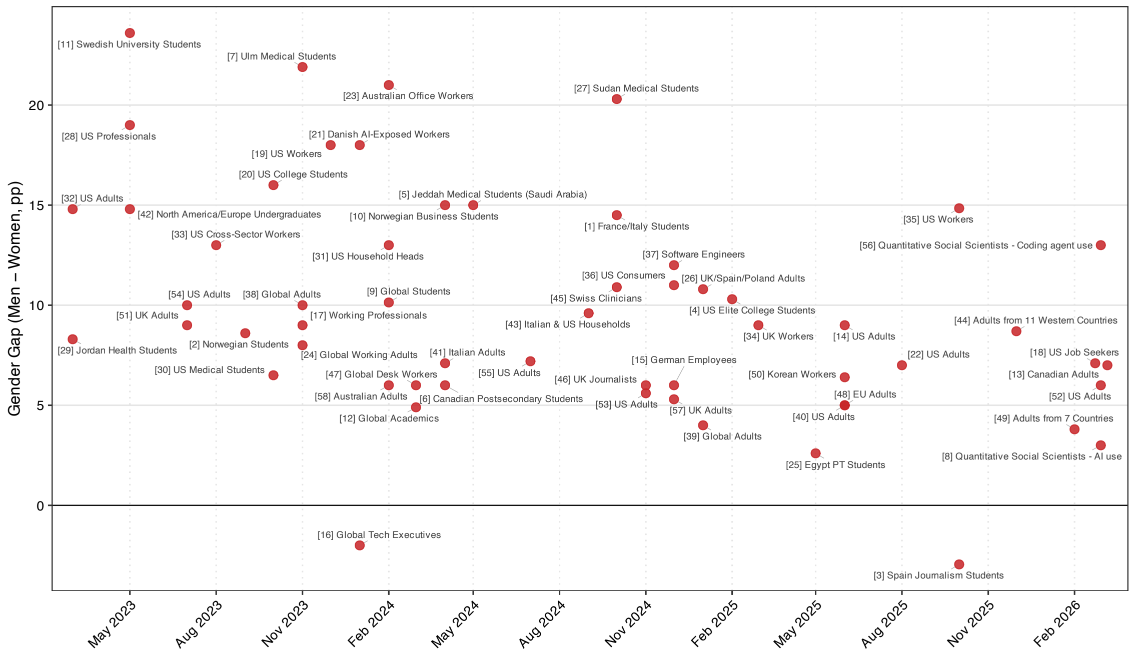
Zhang, Janice S., Christine Yoon, Darnell K. Adrian Williams, and Adi Pinkas. 2024. "Exploring the Usage of ChatGPT Among Medical Students in the United States." *Journal of Medical Education and Curricular Development* 11:1–7.

Figure 1: Generative AI Adoption by Gender: Systematic Review



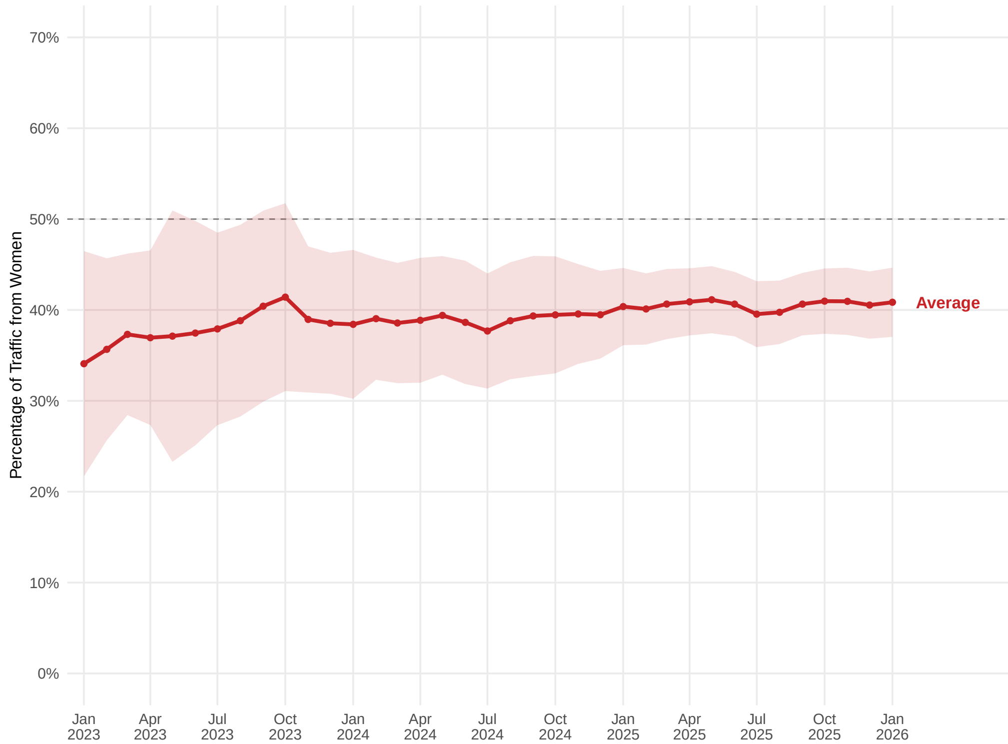
Each row represents one source in our systematic review, showing the percentage of women (red triangles) and men (black circles) in the source's sample who report using generative AI. Sources are ordered from highest to lowest male adoption rate. Each row label gives the source's short name, the first author, the total sample size, and the last month-year of data collection. Sample: sources that report, or allow us to derive, comparable binary adoption rates for both men and women; see Appendix A for figure inclusion criteria.

Figure 2: Gender Gap in Generative AI Adoption Over Time, Across Figure Sources




Each red point represents one figure source in our systematic review, plotted at the time of its data collection (horizontal axis). The vertical axis is the source-level gender gap in generative AI adoption, computed as the share of men who report using AI minus the share of women who report using AI, expressed in percentage points. Positive values indicate higher adoption among men. For sources whose data collection spans multiple months, we assign the source to the final month of that window. Point labels use the bracketed source numbering from Figure 1.

Figure 3: Share of AI Tool Website Traffic from Women Over Time (Worldwide)

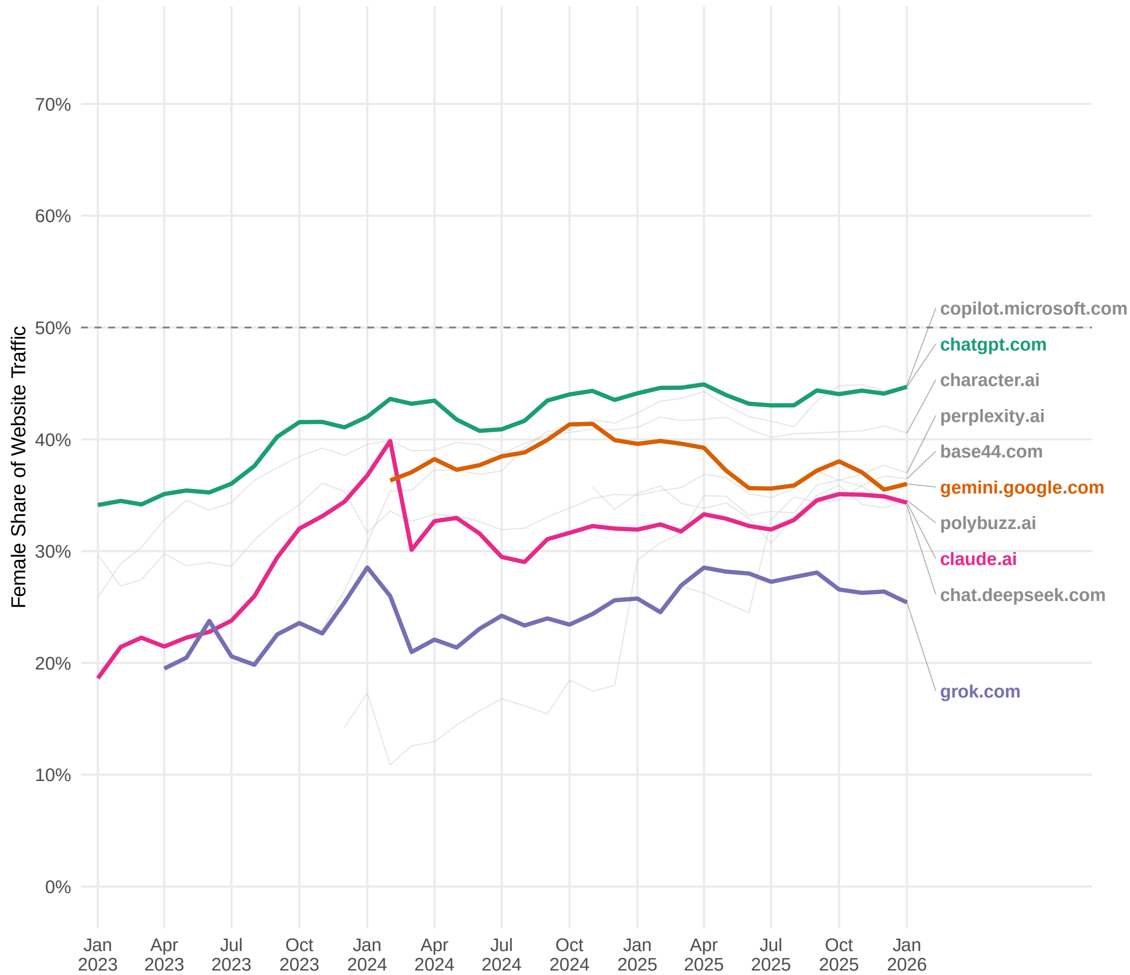



We plot the share of website traffic from female users across the ten highest-traffic domains in SimilarWeb's AI Chatbots and Tools category, observed monthly worldwide from January 2023 through January 2026 (see subsection 3.2 for sample construction). For each domain-month, the female share is constructed as the sum of SimilarWeb's worldwide demographic shares across all age groups for female users; the raw SimilarWeb data are at the (domain, country, month, gender, age) level and are aggregated across age bins within gender. The red line is the unweighted mean across domains. The shaded red band denotes a 95% confidence interval around the cross-domain mean within each month. The dashed horizontal line marks gender parity (50%).

Figure 4: Gender Gap in AI Tool Traffic by Country, August 2025 – January 2026

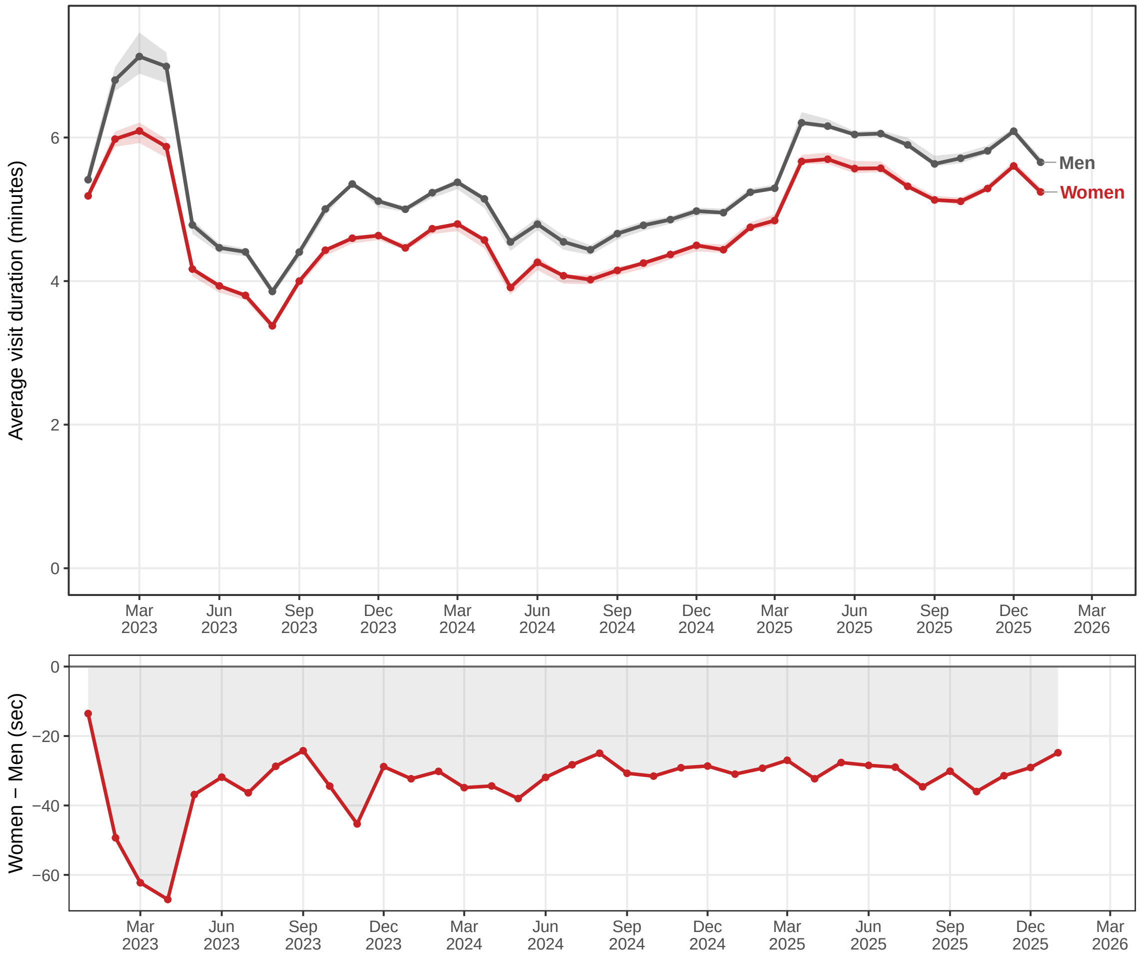
Each row displays the mean female share of AI chatbot website traffic for one country, connected to the 50% parity line by a dark-gray bar. For each domain–country–month, we sum SimilarWeb's female demographic shares across age groups; we then average across the six months from August 2025 to January 2026 within each domain–country pair, and finally average across the ten domains selected by worldwide traffic within each country (see subsection 3.2). Countries are ordered by female share. The row labeled "Global" reports the worldwide aggregate. The dashed vertical line marks gender parity.

Figure 5: Female Share of U.S. AI Tool Website Traffic Over Time, by Tool




Each line plots the female share of U.S. website traffic for a single AI chatbot tool over time, where for each tool-month the female share is the sum of SimilarWeb's U.S. female demographic shares across age groups. The sample is the ten tools with the largest average monthly U.S. visits from August 2025 to January 2026; `claude.ai` and `anthropic.com` are pooled. The four most visited tools in that time window are plotted in color, and all ten tools are labeled at the right edge (see subsection 3.2). The dashed horizontal line marks gender parity (50%).

Figure 6: Average visit duration on chatgpt.com by gender, United States (Jan. 2023 — Jan. 2026).




*Notes:* Data are from our top-ten panel of SimilarWeb's AI Chatbots and Tools category (see subsection 3.2), restricted to `chatgpt.com` and the United States. Top panel: the solid red (women) and gray (men) lines plot the monthly, age-weighted mean visit duration in the U.S., where the six age-bucket durations; 18 — 24, 25 — 34, 35 — 44, 45 — 54, 55 — 64, 65+) are combined using the within-month age × gender share of traffic as weights and converted from seconds to minutes. Shaded bands show the inter-quartile range of raw duration across the six age buckets within each gender-month. The bottom panel plots in red the women-minus-men difference in the age-weighted mean duration shown in the top panel in seconds. Negative (positive) values indicate months in which U.S. men (women) spent more time per visit on chatgpt.com on average.

Table 1: Summary statistics for the top 10 AI tools in the United States.

| | Visits & gender composition | | | | | Visit duration (min.) | | | |
|---|---|---|---|---|---|---|---|---|---|
| Tool | Avg. visits (M) | % total | % Women | % Men | $p$ | Avg. | Women | Men | $p$ |
| chatgpt.com | 1,056.4 | 70.45 | 44.1 | 55.9 | <0.001 | 5.57 | 5.28 | 5.80 | <0.001 |
| gemini.google.com | 171.7 | 11.45 | 36.6 | 63.4 | <0.001 | 5.93 | 5.51 | 6.17 | <0.001 |
| grok.com | 57.9 | 3.86 | 26.7 | 73.3 | <0.001 | 6.71 | 5.89 | 7.00 | 0.005 |
| claude.ai | 55.5 | 3.70 | 34.5 | 65.5 | <0.001 | 5.17 | 5.07 | 5.23 | 0.194 |
| character.ai | 43.8 | 2.92 | 40.7 | 59.3 | <0.001 | 17.80 | 18.79 | 17.12 | <0.001 |
| perplexity.ai | 32.0 | 2.13 | 36.8 | 63.2 | <0.001 | 4.55 | 4.13 | 4.80 | <0.001 |
| copilot.microsoft.com | 18.3 | 1.22 | 43.9 | 56.1 | <0.001 | 5.88 | 5.72 | 6.01 | <0.001 |
| chat.deepseek.com | 18.0 | 1.20 | 35.0 | 65.0 | <0.001 | 4.71 | 4.43 | 4.86 | 0.004 |
| polybuzz.ai | 14.8 | 0.99 | 34.4 | 65.6 | <0.001 | 19.84 | 21.30 | 19.08 | <0.001 |
| base44.com | 4.0 | 0.27 | 35.5 | 64.5 | <0.001 | 9.68 | 9.39 | 9.83 | 0.201 |

Notes: Sample is the top 10 AI chatbot tools by average monthly US visits from August 2025 to January 2026; `claude.ai` and `anthropic.com` are pooled. Average visits (column 1) are in millions, and "% total" is each tool's share of total average monthly US visits across the SimilarWeb panel. Gender shares and visit duration are monthly averages over the same window. Visit duration is reported in minutes. $p$-values are paired $t$-tests comparing monthly women and men shares and durations.

Table 2: Mechanisms Proposed in the Literature: Summary and Supporting Sources

| Adoption friction | Summary and supporting sources |
|---|---|
| **1. Knowledge and familiarity** | Women report lower familiarity and self-assessed knowledge of generative AI (Aldasoro et al., 2024; Stöhr, Ou, and Malmström, 2024; Draxler et al., 2023), score lower on objective LLM knowledge tests (Egli et al., 2025), and more often cite lack of knowledge or not knowing how to use AI as a barrier (Humlum and Vestergaard, 2025; Chugunova et al., 2026). In Aldasoro et al. (2024), self-assessed GenAI knowledge explains most of the raw gender gap in use, while Draxler et al. (2023) show that the gap narrows substantially among respondents with technology-related educational backgrounds. In the job-search setting, Bachaud (2026) finds that women use AI less than men across every job-search application. |
| **2. Perceived usefulness** | Women often report lower or less certain expected returns from GenAI. They perceive smaller job-opportunity benefits in Aldasoro et al. (2024), lower educational usefulness in Carvajal, Franco, and Isaksson (2024) and Stöhr, Ou, and Malmström (2024), and lower expected productivity gains in workplace surveys (Arbanas et al., 2024; Lean In, 2026). Stone (2025) also finds that male students report greater excitement about AI. Angrisani et al. (2026) find that women are less likely to use LLMs for curiosity or entertainment. However, Humlum and Vestergaard (2025) find no gender gap in average expected time savings, but show that women have greater uncertainty about time savings. Borwein et al. (2026) find that women's lower support for AI adoption narrows when AI is described as more certain to produce positive labor-market effects (more jobs, higher wages, etc). |
| **3. Institutional support, training, and confidence** | Women report weaker employer or manager encouragement and less workplace training on how to use AI (Arbanas et al., 2024; Lean In, 2026). They also systematically report lower confidence using AI tools. Barisano et al. (2024) find lower AI-skill confidence and lower willingness to use AI without clear company policy. Carvajal, Franco, and Isaksson (2024) finds lower prompt confidence and lower persistence when prompts are unsuccessful. Chugunova et al. (2026) finds less engagement with prompt-engineering resources among female researchers. Monday Girl (2026) similarly reports lower AI confidence and comfort among women, alongside higher confusion and intimidation. Humlum and Vestergaard (2025) find that women are more likely to report needing training to use AI before benefiting, and Public First and Google (2025) finds that short hands-on training substantially increases women's AI use. Women also are especially responsive to institutional restrictions, reducing intended or reported AI use more than men when AI is banned, unclear rules, or regulated (Carvajal, Franco, and Isaksson, 2024; Contractor and Reyes, 2025; Chugunova et al., 2026). |
| **4. Social legitimacy and status costs** | Women face and anticipate stronger social penalties from AI use. In the workplace, women are more likely to worry that AI use will be perceived as cheating or have negative consequences (Lean In, 2026; Monday Girl, 2026). In student settings, men are more willing than women to use AI in ambiguous or normatively contested settings, and women reduce intended use more sharply when AI is explicitly banned (Stone, 2025; Carvajal, Franco, and Isaksson, 2024; Contractor and Reyes, 2025). Gai, Hou, and Tu (2025) show that identical AI-assisted work receives lower competence ratings, with a substantially larger penalty for female engineers than for male engineers, and that female engineers anticipate stronger managerial penalties for using AI. |
| **5. Trust, privacy, and risk concerns** | Women report lower trust that GenAI providers will keep their data secure and higher privacy concerns (Arbanas et al., 2024; Aldasoro et al., 2024); concerns about AI's effects on mental health, privacy, climate, and the labor market are stronger predictors of women's non-use (Stephany and Duszynski, 2026); and women are more likely to judge AI's risks as exceeding its benefits (Borwein et al., 2026). Lean In (2026) finds that women are more likely to question AI accuracy, while Carvajal, Franco, and Isaksson (2024), Barisano et al. (2024), and Chugunova et al. (2026) find little or no evidence that women are generally less trusting of AI accuracy or returns. Some evidence shows that women do not appear less likely than men to use AI because they personally fear losing their jobs (Humlum and Vestergaard, 2025; Howington, 2024), but there is some evidence of gendered concerns that AI may be more likely displace women as a group (Lean In, 2026; Borwein et al., 2026). |

# Supporting Information

## Global Evidence on Gender Gaps and Generative AI

### Table of Contents

# A Overview of Systematic Review Data

For our systematic review, to find sources that would allow us to look at gender differences in generative AI use, we conducted a comprehensive search on Google Scholar, Scopus, and open internet sources in May 2026 looking for any empirical evidence on gender and generative-AI engagement. We retained sources if they studied a named generative-AI tool or an explicitly defined class of generative-AI tools, reported respondent, worker, student, author, or visitor gender, and provided original quantitative evidence on adoption, frequency, intensity, audience composition. We did not restrict eligibility by geography, sector, publication status, or whether the source appeared in a peer-reviewed outlet. Our sources span academic pre-prints, peer reviewed articles, industry articles, and government reports. Each source in our systematic review is listed below.

The broader systematic review corpus includes 76 sources, while the pooled adoption figures use 58 plotted source-outcome entries. For an entry to be included in Figures 1 and 2, it had to directly report, or allow us to derive from a published table, figure, or underlying data, one male adoption rate and one female adoption rate measured on the same sample and the same AI-use outcome. If a source reported multiple gender AI use measures, we selected the measure closest to broad adoption. We document all measurement choices in Table A2. 18 sources are retained in the broader narrative review, but are not plotted in these figures, as these sources report only regression coefficients, odds ratios, qualitative direction-of-effect statements, gender as a control in a decomposition, or non AI use outcomes such as attitudes, anxiety, familiarity, or trust in AI without male and female adoption rates. The sample-size-weighted average reported in the text uses the 54 plotted source-outcome entries with exact numeric analytic sample sizes, which together cover 318,924 respondents or observed users.[12]

Several sources of potential bias should be kept in mind when interpreting this systematic review. First, the corpus may be affected by publication and reporting bias: studies, industry reports, and public-facing charts that find large or otherwise salient gender differences may be more likely to report them, while null gender subgroup results may remain in unpublished cross-tabs or appendices. At the same time, most sources in the corpus were not designed primarily to study gender. In many cases, the gender comparison is a descriptive subgroup result from a broader survey of AI adoption. This mitigates the concern that every source entered the literature because it was written around a significant gender gap, but it also means that gender measures, subgroup sample sizes, and reporting conventions are inconsistent across studies. Second, sampling frames vary widely. Some studies use nationally representative or quota samples, while others rely on convenience samples, professional mailing lists, student samples, platform users, or industry panels. In the table below, we indicate if a source is nationally representative. Third, because many reports do not publish their exact survey language, our ability to verify exact question wording is limited.

In Table A1, we report each source's sample size, sample description, source type, and data-collection dates. In Table A2, we report the verbatim survey-question wording for the plotted gender estimate where available, or the closest published description of the measure when exact wording was not available. For each plotted entry, we extracted the male and female adoption rates, denominator, field period, outcome definition, and coding notes, and checked these values against the source PDF, report text, public table, figure, and/or underlying data (where available).

[12]Public First UK Workers and Eurostat are plotted but excluded from the weighted average because they do not report numerical analytic sample sizes for the plotted gender comparison. Salesforce/YouGov and ZipRecruiter are also excluded from the exact-sample-size weighted calculation because they report approximate sample sizes rather than exact analytic sample sizes.

Table A1: Sources included in the systematic review of the gender gap in generative-AI adoption.

| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| *Panel A. Sources plotted in Figure 1 (ordered by male adoption rate, descending)* | | | | | |
| 1 | Sublime and Renna (2024) | 385 | Higher-education students in France and Italy; gender comparison excludes non-binary/prefer-not-to-say respondents | Academic | May — October 2024 |
| 2 | Møgelvang et al. (2024) | 2,692 | Norwegian higher-education students at a large university college | Academic | September 2023 |
| 3 | Alonso-Muñoz and Casero-Ripollés (2026) | 281 | Journalism students at three Spanish universities | Academic | October 2025 |
| 4 | Contractor and Reyes (2025) | 634 | Undergraduates at Middlebury, a selective US liberal-arts college | Academic | December 2024 — February 2025 |
| 5 | Alammari et al. (2025) | 354 | Medical students in Jeddah, Saudi Arabia, who had heard of ChatGPT | Academic | March — May 2024 |
| 6 | Pizarro Milian and Peters (2024) | 877 | Canadian postsecondary students in Academica's StudentVu 2024 survey; plotted from 2024 wave figure (methodological-note final-analysis sample $n = 798$) | Industry | March 13 — April 21, 2024 |
| 7 | Maaß et al. (2025) | 200 | Medical students at Ulm University; full survey $N = 207$ | Academic | November 2023 |
| 8 | Lyttelton, Massenkoff, and Wilmers (2026) | 935 | Active quantitative social scientists recruited by email for a study on workflows and AI use; AI-use row uses the Figure 3 name-classified gender subgroup (typically male $n = 561$, typically female $n = 374$; full survey $N = 1{,}260$) | Industry | February — March 2026 |
| 9 | Ravšelj et al. (2025) | 21,990 | Higher-education students across 109 countries/territories; binary-gender respondents with nonmissing Q13 (full dataset $N = 23{,}218$) | Academic | October 2023 — February 2024 |
| 10 | Carvajal, Franco, and Isaksson (2024) | 595 | Bachelor's and master's students at NHH Norwegian School of Economics | Academic | November 2023 — April 2024 |
| 11 | Stöhr, Ou, and Malmström (2024) | 5,574 | Students at Swedish universities, recruited via student networks and social media; ChatGPT-by-gender analytic sample (full survey $N = 5{,}894$) | Academic | April — May 2023 |

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| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 12 | Mohammadi et al. (2026) | 1,652 | Publishing academics across 20 countries, drawn from Scopus author records | Academic | January — March 2024 |
| 13 | Monday Girl (2026) | 1,502 | Representative online sample of adult Canadians in the Angus Reid Forum; survey conducted in English and French; exact gender denominators not reported in the public release | Industry | February 24 — 26, 2026 |
| 14 | Kennedy et al. (2025) | 5,023 | Nationally representative US adults, Pew American Trends Panel | Industry | June 9 — 15, 2025 |
| 15 | Arntz et al. (2026) | 9,410 | Nationally representative German socially insured employees aged 16 — 65, DiWaBe 2.0 linked employer — employee survey | Academic | July — December 2024 |
| 16 | Barisano et al. (2024) | 6,558 | Tech-industry employees across four seniority levels and eight functions in India, the US, Japan, the UK, and Germany | Industry | January 2024 |
| 17 | Glassdoor Economic Research (2024) | 5,017 | US working professionals, Glassdoor platform panel | Industry | November 13 — 17, 2023 |
| 18 | Bachaud (2026) | 1,500+ | U.S. job seekers in ZipRecruiter's Q1 2026 Job Seeker Confidence Survey; online sample conducted for ZipRecruiter by PureSpectrum; respondents had to reside in the U.S. and plan to find a new job in the next six months | Industry | February 7 — 23, 2026 |
| 19 | Brooks (2023) | 1,000 | US workers who were familiar with and had used ChatGPT | Industry | 2023 |
| 20 | Nam (2024) | 1,000 | US undergraduate and graduate students, online panel | Industry | September 29 — October 5, 2023 |
| 21 | Humlum and Vestergaard (2025) | 17,907 | Danish workers in 11 AI-exposed occupations, linked to administrative registers; nationally representative within the 11 occupations | Academic | November 2023 — January 2024 |
| 22 | AmeriSpeak and NORC at the University of Chicago (2025) | 1,121 | US adults in the AmeriSpeak Omnibus August 2025 wave | Industry | August 22 — 24, 2025 |
| 23 | Salesforce, Slack, and YouGov (2024) | Over 1,000 | Australian office workers in a Salesforce-commissioned YouGov survey | Industry | Not stated in PDF (released February 27, 2024) |

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| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 24 | Kreacic and Stone (2024) | 25,000 | Workers aged 18 — 65 across 16 countries, Oliver Wyman Forum global survey | Industry | June and November 2023 |
| 25 | El-Sobkey et al. (2026) | 475 | Physical-therapy students at Beni-Suef University in Egypt | Academic | November 2024 — May 2025 |
| 26 | Campus AI (2025) | 1,411 | Adults in the UK, Spain, and Poland (the non-US subset of a four-country survey), census-quota-matched online panel | Industry | December 2024 — January 2025 |
| 27 | Ahmed et al. (2025) | 1,443 | Medical students in Sudan recruited through student groups and medical-student associations | Academic | June 20 — October 30, 2024 |
| 28 | Howington (2024) | 5,641 | US working professionals across generations, education levels, and career levels, FlexJobs panel | Industry | May 3 — 21, 2023 |
| 29 | Sallam et al. (2023) | 109 | Health students in Jordan who had heard of ChatGPT | Academic | February 28 — March 31, 2023 |
| 30 | Zhang et al. (2024) | 126 | Medical students in the United States, binary-gender analytic subgroup | Academic | August — October 2023 |
| 31 | Aldasoro et al. (2024) | 890 | Nationally representative US household heads, NY Fed Survey of Consumer Expectations | Academic | February 2024 |
| 32 | Draxler et al. (2023) | 1,495 | Nationally representative US adults aged 18+ recruited via Prolific, quotas on age, gender, and ethnicity | Academic | March 2023 |
| 33 | Goligoski et al. (2024) | 1,173 | US workers across knowledge, manufacturing, and service sectors, Glimpse non-probability panel | Industry | August 6 — 26, 2023 |
| 34 | Public First and Google (2025) | Not stated | UK workers, Public First landscape poll for Google AI Works (second wave); results reported only by gender × age, plotted value is the unweighted mean across the three age bins (under 35, 35 — 49, 55+) | Industry | February 28 — March 7, 2025 |
| 35 | Hartley et al. (2025) | 3,940 | US workers aged 18+ in the public September/October 2025 US survey microdata; binary-gender respondents after excluding QC-Client rows | Academic | September 24 — October 1, 2025 |
| 36 | Arbanas et al. (2024) | 3,833 | US consumers across age groups, education levels, and industries, Deloitte Connected Consumer Survey | Industry | 2024 |

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| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 37 | Gai, Hou, and Tu (2025) | 28,698 | Full-time software engineers at a global tech company, tracked through server logs | Academic | January — December 2024 |
| 38 | Google and Ipsos (2024) | 3,034 | Adults aged 18+ across 17 countries, Ipsos online panels and KnowledgePanel | Industry | October 19 — November 6, 2023 |
| 39 | Conceição et al. (2025) | 21,000 | Adults across 21 countries, randomized telephone polling covering 63% of the world population | Government | November 2024 — January 2025 |
| 40 | Saad (2025) | 2,017 | Nationally representative US adults, Gallup Work and Education Poll | Industry | June 2025 |
| 41 | Gambacorta, Jappelli, and Oliviero (2025) | 5,005 | Nationally representative Italian adults aged 18 — 75, Bank of Italy Survey of Consumer Expectations | Government | April 2024 |
| 42 | Lopez Barrios and Déri (2025) | 4,470 | Undergraduate students at 31 universities in North America and Europe; binary-gender analytic subgroup | Academic | Approximately May 2023 |
| 43 | Loschiavo et al. (2026) | 3,216 | Nationally representative Italian households (HOS, Household Outlook Survey) and US household heads (SCE), pooled | Government | February, August, September 2024 |
| 44 | English et al. (2026) | 12,765 | Adults in 11 Western countries (Australia, Canada, France, Germany, Italy, Netherlands, Poland, Romania, Spain, UK, US), YouGov nationally representative panels in each country | Industry | September 23 — December 19, 2025 |
| 45 | Egli et al. (2025) | 685 | Clinicians in Switzerland recruited through national medical societies | Academic | April 18 — October 1, 2024 |
| 46 | Thurman, Thäsler-Kordonouri, and Fletcher (2025) | 1,004 | UK journalists across ages, employment types, seniority, reporting beats, and media formats | Academic | August — November 2024 |
| 47 | Sarasohn (2024) | 10,045 | Full-time desk workers in the US, Australia, France, Germany, Japan, and the UK, Slack workforce survey | Industry | March 6 — 22, 2024 |
| 48 | Eurostat (2026) | NA | Individuals in EU household ICT surveys; EU aggregate for AI use in the previous three months | Government | Second quarter 2025 |

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| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 49 | Bick et al. (2026) | 55,415 | Employed adults aged 18 — 67 across seven countries (US, UK, Netherlands, Sweden, France, Germany, Italy), nationally representative within country | Academic | May 21 — June 18, 2025 and January 19 — February 2, 2026 |
| 50 | Suh and Oh (2026) | 5,512 | Nationally representative Korean workers aged 15 — 64, Korea Research web panel | Academic | May 19 — June 17, 2025 |
| 51 | Haslberger, Gingrich, and Bhatia (2023) | 1,041 | UK working-age adults recruited via YouGov, broadly representative of the working-age population with men overrepresented | Academic | July 19 — 28, 2023 |
| 52 | Lean In (2026) | 1,015 | Nationally representative US adults aged 18+ | Industry | March 2 — 6, 2026 |
| 53 | Bick, Blandin, and Deming (2024) | 6,951 | Nationally representative US adults aged 18 — 64, Real-Time Population Survey weighted to CPS | Academic | August and November 2024 |
| 54 | Park and Gelles-Watnick (2023) | 5,057 | Nationally representative US adults, Pew American Trends Panel with address-based recruitment | Industry | July 17 — 23, 2023 |
| 55 | Angrisani et al. (2026) | 8,964 | Nationally representative US adults, Understanding America Study probability-based online panel | Academic | April — July 2024 |
| 56 | Lyttelton, Massenkoff, and Wilmers (2026) | 935 | Active quantitative social scientists recruited by email for a study on workflows and AI use; coding-agent-use row uses the Figure 3 name-classified gender subgroup (typically male $n = 561$, typically female $n = 374$; full survey $N = 1{,}260$) | Industry | February — March 2026 |
| 57 | Stephany and Duszynski (2026) | 8,000 | Nationally representative UK adults, Public Attitudes to Data and AI Tracker | Academic | 2023 — 2024 |
| 58 | Notley et al. (2024) | 3,852 | Nationally representative Australian adults, online panel quota-matched to the population by age, gender, state, and education | Academic | January 24 — February 18, 2024 |

*Panel B. Additional sources in the review corpus not plotted in Figure 1 (alphabetical)*

| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 59 | Björkegren et al. (2025) | 529 | Classroom teachers and principals in Sierra Leone, recruited through government schools and EducAid | Academic | April 2023 — September 2024 |

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| # | Source | N | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 60 | Borwein et al. (2026) | 3,049 | US and Canadian adults, YouGov opt-in panel with post-stratification weights; analytic subset of a 6,056-person survey who answered the generative-AI module | Academic | November 2023 |
| 61 | Chatterji et al. (2025) | ∼1,100,000 | Uniform/random samples of eligible logged-in ChatGPT consumer-plan conversations/accounts (Free, Plus, Pro), with stated exclusions; gender proxied from first names | Academic | May 2024 — June 2025 |
| 62 | Chugunova et al. (2026) | 6,215 | Researchers at the Max Planck and Fraunhofer Societies in Germany | Academic | June 2024 |
| 63 | Dorta-González et al. (2024) | 1,649 | Researchers worldwide, secondary analysis of the Nature "AI and science" survey | Academic | September 2023 (Nature survey) |
| 64 | Gnambs et al. (2025) | 1,098 | German adults aged 18+, probability-based online panel | Academic | Not stated (2024) |
| 65 | Gonzalez Vazquez et al. (2025) | 70,316 | Nationally representative working-age population across all 27 EU Member States, AIM-WORK mobile-phone-based probability sample | Government | 2024 — 2025 |
| 66 | Ibrahim et al. (2025) | 1,007 | German-speaking adults, online convenience panel | Academic | October 2023 — February 2024 |
| 67 | Liu and Wang (2026) | Traffic data | Global users of the top 40 generative-AI tools across 200+ economies, Semrush clickstream estimates | Academic | Monthly traffic, January 2022 — March 2024 |
| 68 | Nordling (2023) | 3,838 | Postdoctoral researchers across 93 countries, Nature's 2023 global postdoc survey | Industry | June — July 2023 |
| 69 | Ofosu-Ampong (2023) | 128 | Higher-education students in Ghana, convenience sample | Academic | 2023 |
| 70 | Pérez-Portabella et al. (2025) | 3,877 | Nationally representative residents of Spain aged 18+, Spanish Center for Sociological Research (CIS) national survey | Academic | February 6 — 15, 2025 |
| 71 | Russo et al. (2025) | 335 | Adults, 98.2% Italian, community convenience sample | Academic | Early 2025 |
| 72 | Shishakly (2025) | 506 | University students across multiple higher-education institutions in the UAE | Academic | November 2024 |

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| # | Source | $N$ | Sample description | Type | Data collection |
|---|---|---|---|---|---|
| 73 | Stone (2025) | 733 | Introductory-psychology undergraduates at Boise State University | Academic | Fall 2023 — Spring 2024 |
| 74 | Tang et al. (2025) | 684,124 / 400 | Study 1: SSRN preprint-author observations, gender inferred from names; Study 2: US researchers who use LLMs, recruited via Qualtrics | Academic | Study 1: May 2022 — June 2023; Study 2: 2024 |
| 75 | Van Noorden and Perkel (2023) | 1,659 | Self-selecting scientists worldwide, Nature “AI and science” survey | Industry | Summer 2023 |
| 76 | Zahs, Schmodde, and Wehner (2026) | 200 / 76 | German employees from diverse industries, Prolific online panel (Study 1: cross-sectional; Study 2: diary) | Academic | Not stated |

*Notes:* Entries 1 — 58 correspond to the source/outcome entries plotted in Figure 1 (numbered in the order they appear in the dumbbell plot, from highest to lowest male adoption). These entries satisfy the figure-level inclusion rule: a male adoption rate and a female adoption rate can be observed or derived for the same sample and the same binary AI-use or AI-adoption outcome. Entries 59 — 76 are additional sources in the review corpus that are not plotted because they do not report directly comparable male/female adoption rates, but they are retained for the narrative synthesis because they provide evidence on gendered AI engagement, use intensity, productivity, attitudes, familiarity, confidence, risk perceptions, or mechanisms. $N$ is the analytic sample on which the reported gender comparison is based. For sources we don't plot, it is the primary analytic sample reported by the source. Source type follows the venue/publisher of each respective article: *Academic* = peer-reviewed journals, conference proceedings, arXiv preprints, and university working papers; *Industry* = company or consultancy reports, trade-press articles, and non-profit research organizations publishing outside peer review (e.g., Pew Research Center, Gallup, Lean In); *Government* = central banks, national statistical offices, and intergovernmental bodies (e.g., BIS, European Commission, UNDP). Samples are labeled as “nationally representative” only where the authors describe the sample as a probability or quota sample designed to represent a defined national (or, where noted, occupational) population. Data-collection dates follow the paper's own reporting. “Not stated” indicates that field dates were not reported.

Table A2: Survey-question wording for each source/outcome entry plotted in Figure 1. Entries are numbered as in the figure (ordered by male adoption rate, descending). Where the cited report reproduces the item stem and response options, we quote them verbatim; where it does not, we indicate so explicitly and quote the closest published characterization of the measure. All row-specific percentages were checked against the cited source PDF or source text used for the extraction.

| # | Source | Survey question or measurement as reported |
|---|---|---|
| 1 | Sublime and Renna (2024) | Respondents were asked question item 3: "Have you used ChatGPT (or an equivalent program) at least once during the course of your studies or in an academic context?" with response options "Yes" and "No." Figure 5 reports ChatGPT use by gender based on this item; after excluding the 10 respondents who either did not answer the gender question or identified as other than male or female, the paper reports 93.8% of male students and 79.3% of female students as users. |
| 2 | Møgelvang et al. (2024) | Respondents were asked "How often do you use/have you used genAI chatbots?" measured on a 5-point scale with options "1 = Daily, 2 = Weekly, 3 = Monthly, 4 = A few times, and 5 = Never." The reported share of users (93.7% of men, 85.1% of women) corresponds to the complement of the "Never" response (options 1 — 4). |
| 3 | Alonso-Muñoz and Casero-Ripollés (2026) | The paper does not reproduce a single binary item stem, but reports the outcome as students who "acknowledged using Gen AI" and states that the prevalence was 91.53% among male students and 94.48% among female students. A frequency table then reports response categories "I never use Gen AI," "I use Gen AI once every six months or less," "I use Gen AI once a month," "I use Gen AI once a week," "I use Gen AI several times a month," "I use Gen AI several times a week," and "I use Gen AI daily." |
| 4 | Contractor and Reyes (2025) | Respondents were asked "Have you ever used any form of Generative AI such as ChatGPT, GitHub Copilot, Claude, etc.?" with response options "Yes" and "No." Users were then asked a frequency follow-up: "How often do you use Generative AI tools during the academic semester?" with options "Never," "Rarely (a few times a semester)," "Occasionally (a few times a month)," "Frequently (a few times a week)," and "Very frequently (daily or almost daily)." The reported "Any use" share (88.7% of men, 78.4% of women) corresponds to any non-"Never" frequency response (Table A1). |
| 5 | Alammari et al. (2025) | Table 3 is labelled "Comparison between students who have used ChatGPT and those who have not used it before as regards socio-demographic data" and compares "Used ChatGPT before: Yes" versus "No." The gender row reports male students 75 (86.2%) "Yes" and 12 (13.8%) "No," and female students 190 (71.2%) "Yes" and 77 (28.8%) "No." These values are conditional on the students included in the usage table (students who had heard of ChatGPT before the study). |
| 6 | Pizarro Milian and Peters (2024) | Figure 1 is captioned "Have you used ChatGPT or other generative AI tools? (2024 n=877, 2023 n=1032)." The text reports that in early 2024, women reported using generative AI at 76% and men at 82%. The methodological note reports that the 2024 survey was fielded from March 13 to April 21, 2024, and that $n = 798$ students were included in the 2024 final analysis after controls; the plotted descriptive percentages use the Figure 1/text denominator of $n = 877$ for the 2024 wave. |
| 7 | Maaß et al. (2025) | The paper does not reproduce the verbatim usage item stem in the main text. In the gender comparison subsection, it reports that among all survey participants, 126 identified as female, "with 60% of them having used ChatGPT," and among the 74 male participants, "significantly more (81.9%) reported ChatGPT use." |

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| # | Source | Survey question or measurement as reported |
|---|---|---|
| 8 | Lyttelton, Massenkoff, and Wilmers (2026) | The article reports the baseline survey asked: “Have you previously used genAI models to aid your research process?” Figure 3 reports this AI-use outcome by name-classified gender: 81% of respondents with typically male names and 78% of respondents with typically female names had previously used genAI models to aid their research process. |
| 9 | Ravšelj et al. (2025) | The Mendeley Data questionnaire prints the screener item in Section 2 (Usage) as Q13: “Have you ever used ChatGPT?” with response options “Yes” and “No.” The questionnaire note immediately below Q13 states: “1 = the questionnaire will continue; 2 = only section 10: Study and personal information will be offered.” The plotted adoption share corresponds to Q13 = “Yes” among binary-gender respondents with a nonmissing Q13 response; using the Mendeley dataset coding, this gives 6,979/(6,979+2,034) = 77.4215% of male students and 8,732/(8,732+4,245) = 67.2852% of female students as users. |
| 10 | Carvajal, Franco, and Isaksson (2024) | Respondents were asked “How familiar are you with generative AI?” with response options “not heard about it,” “heard about it but not using it myself,” “used it a few times,” “use it occasionally,” and “use it all the time.” The reported “regular use” share (75.7% of men, 60.7% of women) corresponds to the combined share choosing “use it occasionally” or “use it all the time.” *Note:* the 75.5% male adoption rate recorded in the meta-analysis data file differs slightly from the 75.7% figure reported in the paper; the paper’s value is used here. |
| 11 | Stöhr, Ou, and Malmström (2024) | Respondents were asked to “rate your familiarity and frequency of use with a selection of AI chatbots” (including ChatGPT), with response options “Unfamiliar; Familiar but never use it; Familiar but rarely use it; and Familiar and regularly use it.” The published paper does not print a gender-specific “ever used ChatGPT” share directly; the 75.4%/51.8% (men/women) split recorded in the meta-analysis data file is derived from the paper’s underlying frequency distributions (sum of “Familiar but rarely use it” and “Familiar and regularly use it”) and should be treated as a derived quantity rather than a number reproduced verbatim by the authors. |
| 12 | Mohammadi et al. (2026) | The authors do not reproduce the verbatim filter question used to determine whether a respondent has ever used generative AI. They state that “all questions used the phrase ‘Chat GPT or other generative AI tools,”’ and report the plotted outcome as whether respondents had ever used ChatGPT or other generative-AI tools. |
| 13 | Monday Girl (2026) | The public release does not reproduce the verbatim item stem or response options. The closest published characterization is: “Six in ten Canadians report using AI in the past year (64%), with men using it more than women (68% vs. 61%).” The survey was conducted February 24—26, 2026 among a representative sample of 1,502 online adult Canadians who were members of the Angus Reid Forum and was administered in English and French. No full questionnaire, gender-specific denominators, or microdata were found publicly. |
| 14 | Kennedy et al. (2025) | The authors do not reproduce the verbatim survey item in the report. The chart label reads “% who say they interact with artificial intelligence (AI). . . ,” with four response columns shown: “Almost constantly/Several times a day,” “About once a day,” “Several times a week,” and “Less often.” The prose summarizes: “62% of U.S. adults say they interact with AI at least several times a week.” The 67% of men / 58% of women figure corresponds to respondents selecting any option at “Several times a week” or more frequent. A separate Topline and Questionnaire document is linked but not included in the report PDF. |

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| # | Source | Survey question or measurement as reported |
|---|---|---|
| 15 | Arntz et al. (2026) | Respondents were asked “How often do you work with AI-supported computer programs?” measured on a five-point Likert scale “ranging from ‘never’ to ‘always,”’ which the authors aggregate in figures to “Never,” “Rarely/Sometimes,” and “Often/Always.” The item was preceded by a definition describing AI as “enabling computer programs and machines to independently perform tasks that would normally require human intelligence.” |
| 16 | Barisano et al. (2024) | The exhibit footnote states: “How frequently do you use any GenAI tool at work? (We considered respondents to have adopted GenAI if they use it more than once a week.)” |
| 17 | Glassdoor Economic Research (2024) | Respondents were asked “Have you used ChatGPT, or other AI Tools, to help you with tasks at work?” with response options “No” (coded as 0) and “Yes” (coded as 1). |
| 18 | Bachaud (2026) | The article does not reproduce a verbatim item stem for the AI-use-in-job-search battery. The closest published characterization is the embedded Datawrapper chart description: “% of female and male job seekers who use AI for ___ in the job search.” The chart reports the categories “Preparing for interviews,” “Drafting cover-letters,” “Searching for jobs,” “Drafting resumes,” “Improving LinkedIn profile,” “Analyzing job descriptions,” “Generating a headshot,” and “Did not use AI in any way.” The plotted adoption share is computed as 100% minus “Did not use AI in any way”: men 100 - 34.4 = 65.6%, women 100 - 41.5 = 58.5%. |
| 19 | Brooks (2023) | The published report does not reproduce a verbatim stand-alone item producing the “percent who have tried ChatGPT” share. The closest published item wording is “How familiar are American workers with ChatGPT?” with response options “Never heard of it,” “Heard of it, but not familiar,” “Somewhat familiar, but have not used it,” and “Very familiar and have used it.” The gender breakdown (65% of men, 47% of women) appears in a separate chart captioned “ChatGPT Awareness and Adoption: By gender — Percent who’ve tried out ChatGPT,” for which the underlying item stem is not printed. |
| 20 | Nam (2024) | The authors do not provide the survey questions. Instead, they state: “A higher percentage of men than women report having used AI tools such as ChatGPT to help complete assignments or exams (64% vs. 48%).” |
| 21 | Humlum and Vestergaard (2025) | The supplementary appendix reproduces the survey questionnaire. The plotted “used ChatGPT for work” outcome is derived from Q7, asked of respondents who had heard of and used ChatGPT: “For what purposes have you used ChatGPT?” with response options “Work only,” “Leisure only,” and “Both work and leisure.” Respondents selecting “Work only” or “Both work and leisure” are counted as having used ChatGPT for work. |
| 22 | AmeriSpeak and NORC at the University of Chicago (2025) | The plotted value uses the personal-activities item from the August 22 — 24, 2025 wave: “How often do you use AI for personal activities in your everyday life?” The gender chart reports men: Daily 17%, Weekly 17%, Occasionally 28%, Never 37%; women: Daily 20%, Weekly 13%, Occasionally 22%, Never 44%. The reported adoption share sums Daily, Weekly, and Occasionally: 62% for men and 55% for women. |

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| # Source | Survey question or measurement as reported |
|---|---|
| 23 Salesforce, Slack, and YouGov (2024) | The article does not reproduce the verbatim survey question or response options. The gender section reports: "Men were found to be leading in adoption, with 61% already using generative AI. Contrastingly, only 40% of women have admitted to embracing the technology." The article reports a YouGov sample of over 1,000 Australian office workers but does not state exact field dates or exact sample size. |
| 24 Kreacic and Stone (2024) | The authors do not reproduce the verbatim survey question. The article paraphrases the outcome as: "59% of male workers aged 18 — 65 around the world say they use generative AI tools at least once a week, while only 51% of women say the same." The accompanying chart caption reads: "Percentages of global workers by age and gender who say they use generative AI at least once a week." |
| 25 El-Sobkey et al. (2026) | The methods specify that students who utilized ChatGPT "must have been using it for at least 1 month." Table 1 compares "ChatGPT users" and "Non-users" among physical therapy students. The sex counts are female: 139 users and 111 non-users (250 total), and male: 131 users and 94 non-users (225 total), giving 58.2% of men and 55.6% of women as ChatGPT users. |
| 26 Campus AI (2025) | The report does not reproduce the verbatim survey question. The methodology section defines the plotted subgroup as "Users — Respondents who reported using generative AI." |
| 27 Ahmed et al. (2025) | The questionnaire section states that participants were asked whether they had heard of ChatGPT before the study and then whether they had used ChatGPT before the study. Table 2 is labelled "Univariate analysis of socio-demographic characteristics of medical students against hearing about ChatGPT and its usage" and reports "Usage of ChatGPT before the study." The gender row reports male students 273 (55.3%) "Yes" and 221 (44.7%) "No," and female students 332 (35.0%) "Yes" and 617 (65.0%) "No." |
| 28 Howington (2024) | The report does not reproduce the survey question or response options. The results are described only at a high level: "FlexJobs polled over 5,600 working professionals between May 3, 2023, and May 21, 2023. Through the responses, men and women shared when and how they utilize artificial intelligence." Reported gender differences partition respondents into four mutually exclusive categories (personal use only; personal and professional use; use at work with permission; use at work without permission). |
| 29 Sallam et al. (2023) | Respondents first received the filter item "Have you heard of ChatGPT before the study?" with "yes" required to proceed to the usage item. The next item was "Have you used ChatGPT before the study?" with response options "Yes" and "No." Among respondents who had heard of ChatGPT, the official JMIR Table 2 reports male 34/63 = 54.0% and female 21/46 = 45.7% had used ChatGPT before the study. The local PDF extraction clips the right-hand usage columns, so the usage counts were checked against the official JMIR article table. |
| 30 Zhang et al. (2024) | The Methods section states that the first set of survey questions assessed whether the respondent "has used ChatGPT in the past and if they have used ChatGPT in their medical studies." The plotted outcome is the medical-studies item. Table 1 reports women: 33 ChatGPT users and 37 non-users (70 total), and men: 30 users and 26 non-users (56 total), giving 53.6% of men and 47.1% of women who had used ChatGPT in medical studies. |

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| # Source | Survey question or measurement as reported |
|---|---|
| 31 Aldasoro et al. (2024) | Respondents were asked “How often have you used artificial intelligence tools (such as ChatGPT, Google Bard, DALL-E, . . .) in the past 12 months?” with response options “Never,” “Less than once a month,” “Once a month,” “Once a week,” and “More than once a week.” |
| 32 Draxler et al. (2023) | The OSF survey instrument first defines large language models as algorithms “trained on a huge corpus of natural (human) language” that can “generate new texts based on input prompts, summarize or expand texts, and adapt the style or tone of a text.” Examples shown to respondents were “GPT-3, chatGPT, GPT-4, Bing with GPT integration, Bard, BERT, and tools such as LAIKA and Typing Mind.” The plotted adoption outcome corresponds to the item “Have you ever used a large language model?” with response options “No” and “Yes.” |
| 33 Goligoski et al. (2024) | Respondents were asked “Do you currently use generative artificial intelligence (AI) in your job?” with response options “Yes, as dictated by my organization”; “Yes, for my individual use but not as dictated by my organization”; “Yes, some in alignment with organizational policies and some for my individual use”; “Other (specify)”; “I’m not sure”; and “No” (the source list uses “f. No” with no trailing period on the option itself). The item was preceded by the preamble: “The following questions will ask your thoughts about generative artificial intelligence (AI). These techniques are used to generate new content, such as text, images, and audio. Commonly used examples of generative AI include ChatGPT, Bard, and Photoshop Generative Fill.” |
| 34 Public First and Google (2025) | The report does not reproduce a verbatim survey item. The closest published characterization is the chart label on p. 14: “Percentage who have used generative AI tools at work,” presented for six gender × age cells (Women under 35, Men under 35, Women 35 — 49, Men 35 — 49, Women 55+, Men 55+) and sourced as “Public First, landscape survey, Feb. 28, 2025 — March 7, 2025.” The 46.3%/37.3% (men/women) values plotted in Figure 1 are the unweighted means of visual readings of the three age-bin bars (men: ≈70/46/23%; women: ≈55/37/20%) and should be treated as approximations rather than exact values. |
| 35 Hartley et al. (2025) | The plotted value uses Hartley et al.’s public US Survey September 2025 microdata, fielded September 24 — October 1, 2025. The plotted outcome corresponds to Q14 in the public IncQuery instrument: “Do you use generative AI for your job?” with response options “Yes” and “No.” Among binary-gender respondents after excluding QC-Client rows, 851/1,855 men = 45.876% and 647/2,085 women = 31.031% answered “Yes” to Q14. |
| 36 Arbanas et al. (2024) | The authors do not reproduce the verbatim survey question. The closest published description of the measure is: “These 2024 and 2023 Connected Consumer Surveys collected data on consumers’ reported adoption of generative AI, including experimentation and use for projects and tasks (beyond experimentation).” Results are phrased as: “Thirty-three percent of women surveyed reported using or experimenting with gen AI, vs. 44% of men.” |
| 37 Gai, Hou, and Tu (2025) | Not a survey item. Adoption is measured from server-side digital trace data: the authors classify “engineers as AI adopters if they had used the tool at least once according to the digital trace,” where the firm’s servers “recorded all interactions with the AI tool, including the number of prompts sent.” |

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| # Source | Survey question or measurement as reported |
|---|---|
| 38 Google and Ipsos (2024) | Respondents were asked “In the last twelve months, have you used any of the following?” including the item “An AI application such as ChatGPT or Bard.” The reported gender breakdown is: “43% of men across countries have used an AI application in the last 12 months, compared to 33% of women.” |
| 39 Conceição et al. (2025) | The Human Development Report does not reproduce a single verbatim survey item for the pooled “AI user” headline (“37 percent of women are AI users, compared with 41 percent of men”). The closest published item wording relates to a 30-day measure: “In the past 30 days, have you ever interacted with artificial intelligence, such as chatbots, in any of the following ways?” with response options that include “educational platforms of learning apps,” “health care services or applications,” and “work-related tools or software.” The full 19-question instrument is not reproduced in the report. |
| 40 Saad (2025) | Respondents were asked “How often, if ever, do you use the following AI-powered technologies to assist in your daily life and work?” with the item “Generative AI tools like Chat GPT, Microsoft Copilot or Google Gemini that can answer complex questions and perform complex writing and design tasks” and response options “Daily,” “At least weekly,” “Less often,” and “Never.” |
| 41 Gambacorta, Jappelli, and Oliviero (2025) | Appendix A (Survey questions), item H8: “In the last 12 months, how often have you used an artificial intelligence tool (such as ChatGPT or Gemini)? (one answer only)” with response options “1. Never / 2. Less than once a month / 3. Once a month / 4. Once a week / 5. More than once a week.” The reported adoption share corresponds to respondents selecting any option other than “Never.” |
| 42 Lopez Barrios and Déri (2025) | The survey item reads: “To help me with writing assignments, I use artificial intelligence tools (ChatGPT, AI-Writer, Quillbot, etc.).” Response options in Table 3 are “Never,” “Rarely,” “Sometimes,” “Frequently,” “Almost always or always,” and “Missing.” The plotted adoption share sums all non-“Never” use categories, using the table’s gender denominators: men 13.2 + 14.5 + 8.3 + 3.0 = 39.0%; women 10.4 + 8.7 + 3.5 + 1.6 = 24.2%. |
| 43 Loschiavo et al. (2026) | The plotted pooled “generic use” measure uses both country-specific frequency items. US Survey of Consumer Expectations (Appendix B, Q1new): “How often have you used artificial intelligence tools (such as ChatGPT, Google Bard, DALL-E, . . .) in the past 12 months?” with response options “1 Never / 2 Less than once a month / 3 Once a month / 4 Once a week / 5 More than once a week.” Italian Household Outlook Survey (Appendix C, AIFREQ, asked only if AIKNOW > 1): “How often have you used GenAI tools in the past 12 months?” with response options “1 Never used / 2 Less than once a month / 3 Once a month / 4 Once a week / 5 More than once a week.” The plotted “generic use” share treats any response other than “Never”/“Never used” as use. |
| 44 English et al. (2026) | The report does not reproduce the verbatim survey question. The measure is paraphrased throughout as respondents who “use AI at least once a week for leisure or personal activities” or “use AI in each of these instances at least once a week.” The closest published descriptor is the chart label: “% of adults who told YouGov they use AI at least once a week for leisure or personal activites [sic].” |

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| # | Source | Survey question or measurement as reported |
|---|---|---|
| 45 | Egli et al. (2025) | Online supplemental file 3 reproduces the complete questionnaire. The plotted outcome corresponds to Q1 under "Factors and Outcomes": "How often do you use LLMs (in general)?" with response options "Several times a day," "Daily," "Weekly," "Monthly," "Fewer than monthly," and "Never." Frequent use is defined as at least weekly use, corresponding to "Several times a day," "Daily," or "Weekly." |
| 46 | Thurman, Thäsler-Kordonouri, and Fletcher (2025) | Respondents were asked (item "gen_per_use"): "How frequently do you personally work with AI in your journalistic tasks?" with response options "Daily," "1 — 4 times a week," "1 — 3 times a month," "Less than once a month," and "Never" (Figure 5). The plotted adoption share corresponds to at least weekly professional use: "Daily" or "1 — 4 times a week." |
| 47 | Sarasohn (2024) | The report does not reproduce the verbatim survey question. The measure is paraphrased variously as respondents who have "tried AI tools," "tried AI for work," "tried AI tools at work," or "experimented with AI tools." The reported gender breakdown reads: "more men trying AI for work (35% of respondents) compared with women (29% of respondents)." |
| 48 | Eurostat (2026) | The article reports EU survey data on "Use of generative AI tools by population group" and states that in 2025, 33% of people aged 16 — 74 living in the EU had used generative AI tools in the previous 3 months. It further reports that men used generative AI slightly more than women (35% vs 30%). The data source note says the statistics come from the EU survey on ICT in households and by individuals; in most EU countries, surveys are carried out in the second quarter asking about people's activities in the previous 3 months. |
| 49 | Bick et al. (2026) | Respondents were asked "Do you use Generative AI for your job?" with response options "No" and "Yes." The question was preceded by the definition: "Generative AI is a type of artificial intelligence that creates text, images, audio, or video in response to prompts. Some examples of Generative AI include ChatGPT, Gemini, and Midjourney." |
| 50 | Suh and Oh (2026) | The authors do not reproduce the verbatim survey question. The plotted gender split in the meta-analysis data file corresponds to the "used at least 1 day last week" segment shown in Figure 2(a). Section 2.3 defines this measure as "Used last week," identifying active users who utilized GenAI in the week preceding the survey. |
| 51 | Haslberger, Gingrich, and Bhatia (2023) | The paper does not reproduce the verbatim stem of the pre-task frequency item from which the 33%/24% "frequent user" share is derived. The Table 2 note reads: "Respondents are classified as frequent users if they report using ChatGPT at least once per week." |
| 52 | Lean In (2026) | The report does not reproduce the verbatim survey question. The measure is characterized only as: "Men are 22% more likely to use AI daily or constantly at their jobs (33% of men vs. 27% of women)." The methodology note states that "LeanIn.Org surveyed a total of 1,015 nationally representative Americans ages 18+ from March 2 — 6, conducted online through Wired Research," but does not reproduce the item wording or response options. |

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| # | Source | Survey question or measurement as reported |
|---|---|---|
| 53 | Bick, Blandin, and Deming (2024) | Employed respondents were asked "Do you use Generative AI for your job?" with response options "No" and "Yes." The question was preceded by the definition: "Generative AI is a type of artificial intelligence that creates text, images, audio, or video in response to prompts. Some examples of Generative AI include ChatGPT, Gemini, and Midjourney." The 29.1%/23.5% (men/women) figures plotted here come from the paper's "Adoption Rate At Work" panel (Figure 5c), which is restricted to employed individuals aged 18 — 64 ($N = 6{,}951$) answering "Yes" to this at-work item. |
| 54 | Park and Gelles-Watnick (2023) | The published Pew short report does not reproduce the verbatim survey question. The closest published description is the chart label: "Among U.S. adults who have heard of ChatGPT, % who say they have ever used it," and the prose: "Among those who have heard of ChatGPT, 24% say they have ever used it." The separate Pew Topline questionnaire is not included in the published article text. |
| 55 | Angrisani et al. (2026) | Respondents were asked "Have you heard about or used AI applications that create human-like text or code, such as ChatGPT, Bard/Gemini, or Bing Chat?" with response options "I have never heard about them," "I have heard about them but never used one," "I have used them," and "I don't know." The reported "ever used LLMs" share (27.2% of men, 20.0% of women) corresponds to respondents who selected "I have used them." |
| 56 | Lyttelton, Massenkoff, and Wilmers (2026) | The article reports the baseline survey asked: "Do you regularly (more than once a week) use an AI coding assistant integrated into your command line (such as Codex, Cursor, or Claude Code)?" A follow-up question verified that respondents used one of those tools, or Google Antigravity. Figure 3 reports this regular coding-agent-use outcome by name-classified gender: 22% of respondents with typically male names and 9% of respondents with typically female names used coding agents. |
| 57 | Stephany and Duszynski (2026) | The paper does not reproduce the verbatim survey question. The outcome is described as "using GenAI tools frequently — at least once a week — in a personal context," with the data drawn from "the 2023 and 2024 waves of the Public Attitudes to Data and AI Tracker (UK Department for Science, Innovation & Technology)." The original DSIT Tracker item wording is not reproduced in the paper. |
| 58 | Notley et al. (2024) | Respondents were asked "To what extent are you familiar with the following services which make use of AI?" for each of several categories, including "Text-focused generative AI (e.g. ChatGPT, Bard)." Response options: "Not at all," "Unsure/don't know," "I have heard of this but not used it," "I have experimented or tried it," and "I use this regularly." The 16%/10% gender split (men/women) corresponds specifically to respondents choosing "I use this regularly" for text-focused generative AI ("Men are more likely to be regular users than women (16% versus 10%)"). |

*Notes:* Row numbers match those in Figure **??**, which orders source/outcome entries by the reported male adoption rate from highest to lowest. All quoted text is verbatim from the cited report, PDF, working paper, or official source text used for the extraction. Where the cited source does not publish the exact question and response options, we indicate this explicitly and quote the closest published characterization of the measure.

Table A3: Additional sources not included in the systematic-review figure: one-sentence description of the gender gap and sample.

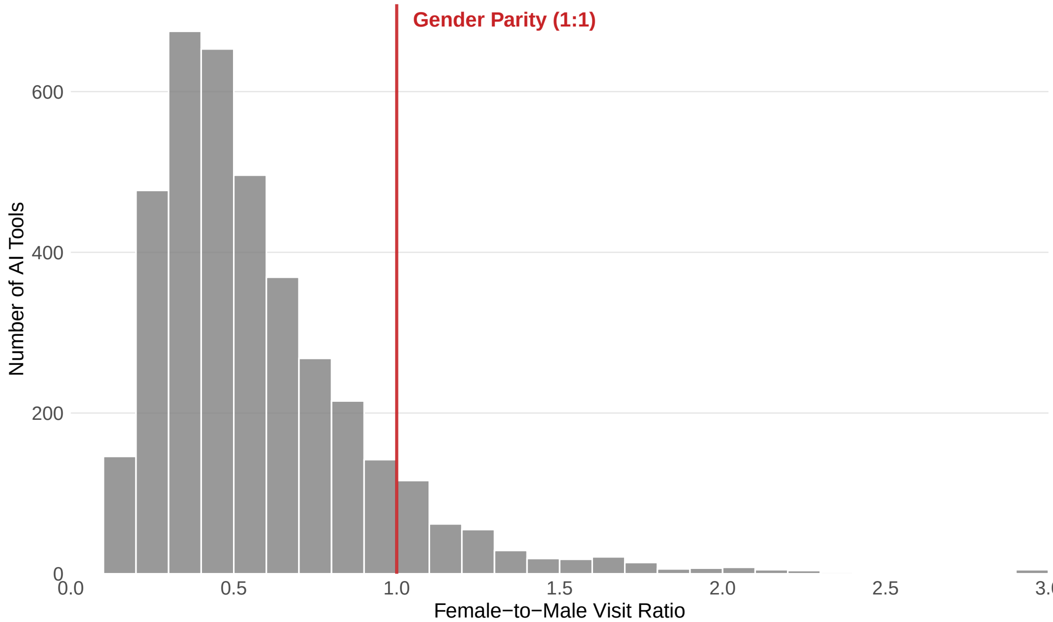
| Source | Description of the gender gap and sample |
|---|---|
| Björkegren et al. (2025) | Female teachers in Sierra Leone were slightly less likely to use a WhatsApp AI chatbot than male teachers over a 17 — month deployment. |
| Borwein et al. (2026) | Among American and Canadian survey respondents, women's agreement that the risks of AI outweigh its benefits was roughly 11% higher than men's. |
| Chugunova et al. (2026) | Female researchers at the Max Planck and Fraunhofer Societies in Germany reported lower familiarity with and less use of AI for research tasks than male researchers, a gap almost entirely explained by observable characteristics. |
| Dorta-González et al. (2024) | In a global survey of 1,659 active scientists drawn from the 2023 Nature AI-and-science poll, female researchers used generative AI tools about 7% less frequently than male researchers (coefficient $-0.067$, $p = 0.022$). |
| Gnambs et al. (2025) | In a random sample of German adults, men reported significantly higher willingness to use AI in the work, healthcare, and education domains than women. |
| Gonzalez Vazquez et al. (2025) | In the AIM-WORK survey of the working-age population across the 27 EU Member States (full sample $N$ = 70,316; regression sample $N$ = 42,433), gender was not a significant factor explaining AI use at work once occupation, education, age, firm size, nationality, income, tenure, and country were controlled for. |
| Ibrahim et al. (2025) | Among German-speaking respondents with prior AI experience, men showed slightly more positive attitudes toward AI and higher perceived ease of use than women, but there were no significant gender differences in perceived usefulness, behavioral intention, or subjective norms around AI. |
| Ofosu-Ampong (2023) | Among higher-education students in Ghana, male students were significantly more likely than female students to use AI-based tools for learning and research. |
| Pérez-Portabella et al. (2025) | In a nationwide CIS survey of 3,877 Spanish adults, being female was negatively and significantly associated with generative-AI use ($\beta = -0.156$, $p = 0.002$) in a linear model that also controlled for performance expectancy, innovativeness, knowledge, social utilities, perceived privacy risk, social risk, need for regulation, generation, education, and income. |
| Russo et al. (2025) | In a community sample of Italian adults, women reported significantly lower prior use of AI than men, alongside higher AI anxiety and less positive attitudes toward AI. |
| Shishakly (2025) | Among undergraduates at six private universities in the United Arab Emirates, female students reported more frequent use of ChatGPT than male students. |
| Stone (2025) | Among undergraduates at a U.S. university, men reported greater familiarity with AI, more self-reported AI usage in both allowed and ambiguous academic contexts, and greater near-term and long-term excitement about AI than women. |
| Tang et al. (2025) | Using SSRN preprint author-month data and a companion survey of U.S.-based academic LLM users, the post-ChatGPT increase in preprint productivity was 6.4% higher for male than female researchers, and in the survey male researchers reported spending more time on and using generative AI more frequently than women. |
| Zahs, Schmodde, and Wehner (2026) | In two samples of German employees, women reported lower intentions to use generative AI at work and used it about 17.5 minutes less per day than men. |
| Chatterji et al. (2025) | In internal ChatGPT consumer-account data, the initially male-skewed user base narrowed over time, with name-inferred feminine users reaching near parity among a sample of consumer accounts by mid-2025. |
| Liu and Wang (2026) | Using Semrush clickstream data on the top 40 generative-AI tools across 200+ economies, the authors document persistent gender differences in AI-tool traffic composition. |
| Nordling (2023) | In a global postdoc survey conducted by Nature, women postdoctoral researchers reported lower use of ChatGPT or similar AI chatbots than men. |
| Van Noorden and Perkel (2023) | In Nature's global AI-and-science survey, women scientists reported lower engagement with generative-AI tools than men. |

# B There's An AI for That Data

As a breadth check on our top-10 AI tools panel, we scraped the webpages of all AI tools listed on *theresanaiforthat.com* (TAAFT) as of April 2025. TAAFT is a public online directory and search platform that lists thousands of AI products and tools across a wide range of use cases, with links to the tool's external website. Using each scraped webpage, we extract the URL for the AI tool itself and use those URLs to query SimilarWeb's database. We scrape 12,976 unique external URLs after accounting for duplication. Out of these, 11,365 domains can be matched to SimilarWeb website visitor data. At the most disaggregated level, the data are cells at the domain–country–month–gender–age level. We aggregate the data across age bins within gender to construct shares of visitors by gender for each domain–country–month. Each tool is assigned a category on TAAFT. Among tools with SimilarWeb traffic data, we observe 2,616 categories. We look more closely at categories with at least 1,000 average monthly visits and more than five unique tools. The traffic filter does not result in any categories being dropped, while the tool-count restriction drops 369 categories. This leaves 128 eligible categories, out of which we highlight the 25 most popular by average monthly visits. Throughout this appendix, we use worldwide traffic only, because the TAAFT panel does not support the cross-country disaggregation from main-text analysis due to size constraints.

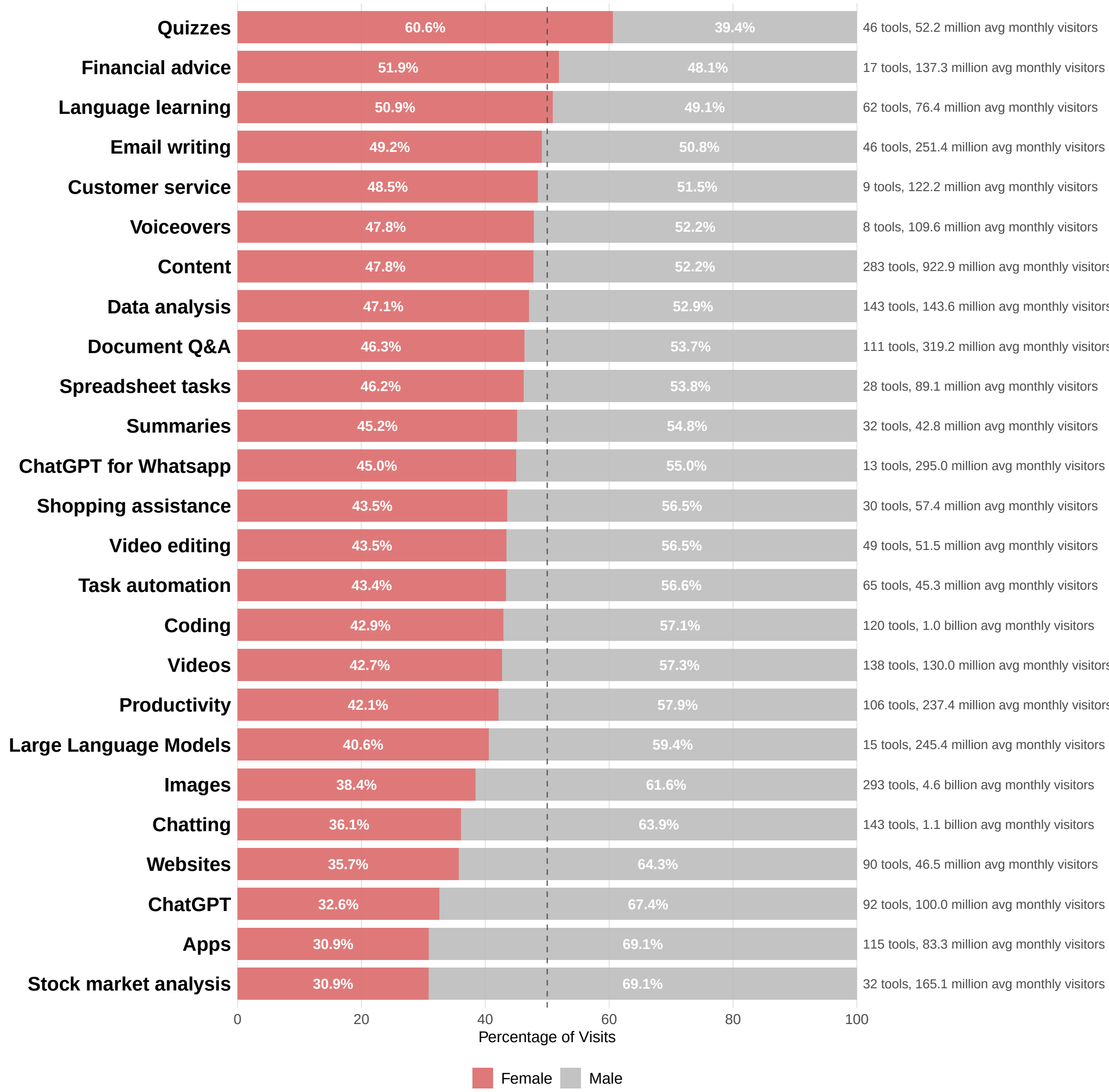
Three descriptive exercises drawn from this broader universe reinforce and extend the patterns we document in the main text. First, Figure A1 plots the distribution of the female-to-male visit ratio across the high-traffic subset of the TAAFT universe (tools with at least 1,000 average monthly visits). The distribution is right-skewed and concentrated well below the parity line at a ratio of one: the large majority of generative-AI tools in the wild attract more male than female visits, consistent with the top-10 panel in the main text but now across thousands of products. Second, Figure A2 shows that this aggregate picture masks substantial category-level heterogeneity. Among the 25 highest-traffic TAAFT categories with more than five tools, categories such as quizzes, learning, meetings, and language learning attract close-to-parity or female-majority traffic, whereas coding, images, image editing, chatting, websites, ChatGPT-wrapper tools, and general "Apps" are heavily male-skewed. Third, to make these category-level patterns tangible at the level of individual products, Figure A3 plots the gender composition of an illustrative set of tools spanning education, writing, image editing, accessibility, finance, and hardware development. Tools oriented toward education (twee, CasperPractice), personal life (GiftList Genie, Maid of Speeches, Baby Face Generator), and accessibility (Be My Eyes) skew female, while tools oriented toward productivity (Moonbeam), analytics (Particl), finance (StockGPT), and hardware engineering (Hestus) skew male.

Figure A1: Distribution of Female-to-Male Visit Ratios Across High-Traffic AI Tools



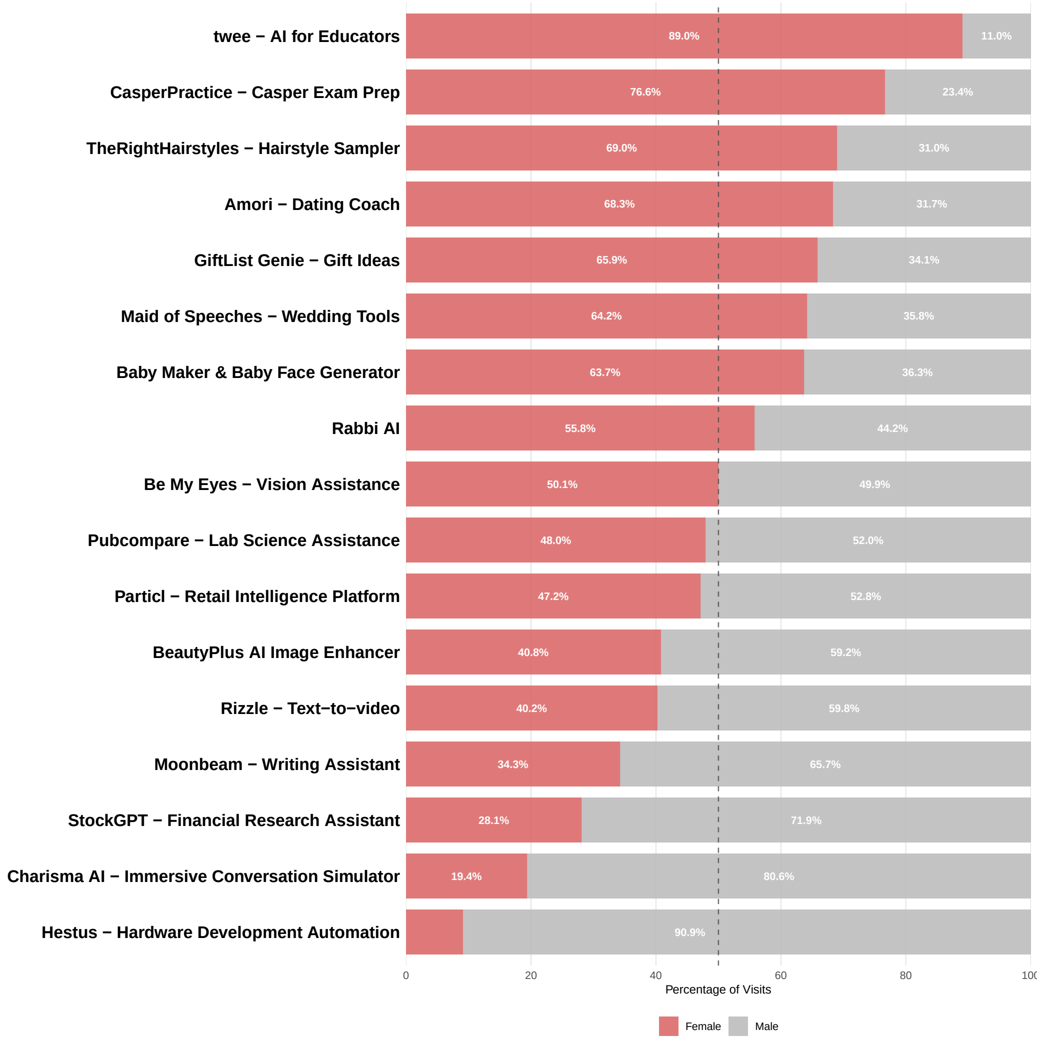

Histogram of the female-to-male visit ratio across generative AI tool websites in the *There's An AI For That* (TAAFT) universe. We scraped all AI tools listed on `theresanaiforthat.com` as of April 2025 (12,976 unique external URLs after accounting for duplication); 11,505 URLs were matched to SimilarWeb traffic and demographics data. For each tool, the ratio is computed as cumulative female visits divided by cumulative male visits, summed across all available months. The sample is restricted to tools with at least 1,000 average monthly visits. Ratios are winsorized at 3.0 for display (values above 3 are set to 3); bin width is 0.1. Values below one indicate male-dominated tools (more male than female visits); values above one indicate female-dominated tools. The solid red vertical line marks gender parity at a 1:1 ratio. Source: SimilarWeb data, August 2022 – July 2025.

Figure A2: Gender Composition of AI Tool Website Traffic by Category




Each bar displays the gender composition of total website traffic for one AI tool category from *There's An AI For That* (TAAFT), with the female share in red on the left and the male share in gray on the right. For each category, gender shares are computed by summing cumulative female and male visitor counts across all tools and months within the category and dividing by their total. Starting from the full TAAFT universe, we scraped 12,976 unique external URLs in April 2025, corresponding to 11,753 parsed domains; 11,365 of which can be merged to SimilarWeb for visitor demographic infomration. After excluding app-store and non-AI domains (such as `play.google.com`, `apps.apple.com`, `google.com`, `facebook.com`, `twitch.tv`, `microsoft.com`, `zoom.us`) and rows without traffic data, the category analysis includes 11,358 unique domains and 2,616 categories. We restrict to categories with at least 1,000 average monthly visits and more than five unique tools, leaving 369 eligible categories; we then display the 25 with the highest average monthly visits. Each row label gives the category name, its unique-tool count, and its average monthly visits from men and women from August 2022 to July 2025. Bars are ordered from highest (top) to lowest (bottom) female share. The dashed vertical line marks gender parity (50%).

Figure A3: Gender Composition of AI Tool Website Traffic for Illustrative Tools




Each bar displays the gender composition of total website traffic for a single AI tool in the *There's An AI For That* (TAAFT) universe, with the female share in red on the left and the male share in gray on the right. Gender shares are computed by summing cumulative female and male visitor counts across all available months for each tool and dividing by their total. The 17 tools shown were chosen to illustrate diverse use cases spanning education and exam preparation (twee, CasperPractice), personal-life assistance (GiftList Genie, Maid of Speeches, Baby Face Generator, Amori), faith (Rabbi AI), image enhancement (BeautyPlus), accessibility (Be My Eyes), scientific and retail analytics (Pubcompare, Particl), writing assistance (Moonbeam), video generation (Rizzle), finance (StockGPT), conversation simulation (Charisma AI), and hardware-development automation (Hestus). All tools shown meet a high-traffic screen of at least 1,000 average monthly visits. Tools are listed in the fixed order above rather than sorted by share. The dashed vertical line marks gender parity (50%). Source: SimilarWeb and TAAFT data, August 2022 – July 2025.

Table A4: Summary Statistics for TAAFT Illustrative AI Tools

**Panel A. Traffic and Gender Composition**

| Tool | Category | Months | Visits | Unique Visitors |
|---|---|---|---|---|
| twee - AI for Educators | Teaching tools | 28 | 262,247 | 91,435 |
| CasperPractice - Casper Exam Prep | Test prep | 5 | 26,236 | 11,386 |
| TheRightHairstyles - Hairstyle Sampler | Hairstyles | 36 | 2,582,181 | 1,776,446 |
| Amori - Dating Coach | Conversation | 2 | 89,110 | 43,941 |
| GiftList Genie - Gift Ideas | Gift ideas | 4 | 78,243 | 31,814 |
| Maid of Speeches - Wedding Tools | Speeches | 12 | 60,222 | 33,486 |
| Baby Maker & Baby Face Generator | Images | 21 | 373,462 | 232,755 |
| Rabbi AI | Religion | 2 | 29,347 | 18,109 |
| Be My Eyes - Vision Assistance | Image-to-text | 31 | 256,106 | 131,717 |
| Pubcompare - Lab Science Assistance | Research | 8 | 102,544 | 53,318 |
| Particl - Retail Intelligence Platform | Comp. analysis | 12 | 82,043 | 33,824 |
| BeautyPlus AI Image Enhancer | Upscaling | 28 | 243,524 | 139,687 |
| Rizzle - Text-to-video | Video | 9 | 29,409 | 13,952 |
| Moonbeam - Writing Assistant | Writing | 12 | 93,830 | 44,852 |
| StockGPT - Financial Research Assistant | Finance | 2 | 133,749 | 69,234 |
| Charisma AI - Immersive Simulator | Characters | 4 | 57,585 | 34,332 |
| Hestus - Hardware Dev Automation | CAD | 1 | 10,676 | 7,117 |

**Panel B. Gender Composition and Engagement Metrics**

| Tool | Female (%) | F/M | Duration | Bounce (%) | Pages |
|---|---|---|---|---|---|
| twee - AI for Educators | 89.0 | 8.12 | 3.30 | 42.9 | 3.72 |
| CasperPractice - Casper Exam Prep | 76.6 | 3.28 | 12.90 | 32.7 | 9.86 |
| TheRightHairstyles - Hairstyle Sampler | 69.0 | 2.23 | 2.70 | 52.3 | 9.86 |
| Amori - Dating Coach | 68.3 | 2.16 | 5.94 | 33.8 | 4.73 |
| GiftList Genie - Gift Ideas | 65.9 | 1.93 | 2.73 | 42.0 | 3.88 |
| Maid of Speeches - Wedding Tools | 64.2 | 1.79 | 1.66 | 57.9 | 1.92 |
| Baby Maker & Baby Face Generator | 63.7 | 1.76 | 3.27 | 36.8 | 3.08 |
| Rabbi AI | 55.8 | 1.26 | 1.31 | 84.5 | 2.20 |
| Be My Eyes - Vision Assistance | 50.1 | 1.00 | 1.35 | 69.0 | 1.81 |
| Pubcompare - Lab Science Assistance | 48.0 | 0.92 | 0.96 | 57.4 | 1.73 |
| Particl - Retail Intelligence Platform | 47.2 | 0.89 | 6.50 | 34.3 | 9.21 |
| BeautyPlus AI Image Enhancer | 40.8 | 0.69 | 1.91 | 66.8 | 2.19 |
| Rizzle - Text-to-video | 40.2 | 0.67 | 9.90 | 32.3 | 11.31 |
| Moonbeam - Writing Assistant | 34.3 | 0.52 | 3.27 | 42.8 | 4.34 |
| StockGPT - Financial Research Assistant | 28.1 | 0.39 | 2.48 | 49.0 | 3.18 |
| Charisma AI - Immersive Simulator | 19.4 | 0.24 | 2.48 | 49.2 | 4.28 |
| Hestus - Hardware Dev Automation | 9.1 | 0.10 | 0.06 | 92.4 | 1.15 |

*Notes:* Summary statistics for illustrative AI tools. Metrics are averaged over months with available data. Female share is based on cumulative visitors. F/M is the female-to-male visitor ratio. Source: SimilarWeb and TAAFT (through July 2025).